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[About the UltraWave 12G Subsystem : Features](#)

Features

The UltraWave 12G subsystem has the following features and capabilities:

- 12GHz source and measure
- Very low phase noise synthesizers and sources
- CW and modulated sources. Individual tones in a multitone can be modulated.
- Receivers have programmable IF filters, including bypass, and two digitizers:
 - High speed (45MHz, 16 bit)
 - High resolution (500kHz-800kHz BW, 24 bit)
- Reference synthesizer provides 4 DUT LO drive ports and 4 DUT clock drive pins
- Universal ports/channels. Each 12GHz MW instrument channel with 1:4 multiplexer has the same features and performance:
 - Independent source level control with level calibration using stabilized reference
 - Step attenuator to maximize source signal integrity and optimize receive SFDR
 - Pin diode switches for rapid settling, high isolation, and low loss
 - Wideband noise source for calibrated ENR
 - Low-noise amplifier for receive path
 - Add path (signal combiner and connection) to a paired channel for multitone source signals
- Scalable architecture provides for high pin count devices and high multisite parallel efficiency

- Configurations range from basic minimum to maximum in incremental steps that add measure boards and synthesizer boards:
 - Minimum: 16 universal ports, with 2 synthesizers (receive LO and one CW/Modulated source) and 2 receivers
 - Full: 96 universal ports, with 8 synthesizers (including 2 Reference sources) and 12 receivers
- High parallel capabilities. For example:
 - A configuration with 64 ports and 4 synthesizers in 6 test head slots provides 16 parallel source ports and 8 parallel measure ports
 - Background data handling and DSP
- Synchronization (SyncLink) with pattern control and pattern sets
- Software “logical instrument” interfaces to the MW hardware functions that simplify programming of subsystem resources for measurements

For hardware specifications, see UltraWave 12G Specifications.

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About the UltraWave 12G Subsystem : Safety Information

Safety Information

When operating instruments using the UltraWave 12G subsystem, always follow general UltraFLEX test system safety guidelines. See [About Operator and Safety Information](#).

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About the UltraWave 12G Subsystem : Key Terms

Key Terms

add path	Signal combiner allowing two source signals to be added to create either a two-tone signal or a multitone signal if one or both signals are from a modulated source.
AWG	Arbitrary Waveform Generator. UltraWave 12G has both an I AWG and a Q AWG for the I/Q modulator for the MWModulatedSource.
board set	A UltraWave 12G instrument requires a minimum of two boards: a synthesizer board and a measure board. Larger configurations have more of each kind. See Supported UltraWave 12G Configurations.
CW	Continuous wave. A signal type equivalent to sine wave.
dBm	Decibel unit of power referenced to 1mWatt into 50ohms.
DSP computer	Computer in the test system support cabinet for DSP computations. A test system can have two DSP computers. Data connect to instruments is MoveLink.
DUT ref and clock channels	Sets of 4 reference LO or clock signals on a MW Synthesizer Reference board and available at the DIB interface. Each channel has an independent disconnect relay.
IF	Intermediate frequency. Frequency at the output of receiver downconverter.
LO	Local oscillator. Used with a mixer to either modulate a carrier or down convert signals.

logical instrument	A set of MW functions configured and connected to perform as a MW instrument with specific capabilities. Examples are MWSource and MWReceiver. Each logical instrument has both VBT language and GUI debugging tool programming interfaces.
MoveLink	High-speed data connection between a receiver digitizer output and a DSP computer.
MW instrument channel or MW channel	An individual low-noise, full-bandwidth signal path in the UltraWave 12G subsystem available to source and measure functions. Each MW channel is “universal” - all have the same features and performance. Contained in a single front-end module on a MW Measure board.
mux	Multiplexer. One-to-many switch allowing a MW instrument channel to be connected to 4 RF I/O ports (tester channels). MW muxes are bidirectional.
MW	Microwave
physical instrument	A set of MW hardware functions configured and connected to perform as a MW logical instrument.
port	An R/F I/O signal connector at the DIB interface providing a signal path to either the DUT or a DIB-mounted application-specific circuit. Each RF port corresponds to a MW tester channel. Compare to a spring-loaded pin.
reference synthesizer	Synthesizer providing reference signals or clocks to a DUT.
RF	Radio frequency
SMPM connector	Blind-mate connector type for each RF I/O port DIB connection.
splitter	Network that splits power across two or more outputs. Allows a single source to drive multiple loads.
VCA	Voltage Controlled Attenuator or Voltage Controlled Amplifier

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About the UltraWave 12G Subsystem : UltraWave 12G Software Licensing Requirements

UltraWave 12G Software Licensing Requirements

UltraWave 12G instruments require one or more licenses to use the following features:

- Arbitrary Waveform Generator
- Noise Source
- Number of channels
- Phase Noise Analyzer
- Source Amplitude Modules
- Vector Network Analyzer

The table below lists licensed features for the UltraWave 12G.

UltraWave 12G Licensed Features

Feature Name	Feature	Comments
8UniversalRFPorts	Universal 8 Port License	Counted license. Each license enables eight ports on a single RF measurement board.

16UniversalRFPorts	Universal 16 (13-16) Port License	Counted license. Each license enables 16 ports on a single RF measurement board.
12GHzReceiver	12GHz Receiver	Counted license. One license is required per receiver operating at 12GHz (default performance is 6GHz).
12GHzSource	12GHz Source	System wide license. If enabled, this license allows 12GHz on all RF ports in the test system.
1PortVNA	Universal Port VNA Option 1 port	Counted per measurement board. Allows for 1-port VNA measurement on any port on the MWMeasure Board.
DUTLOSource	DUT LO Source	Counted license. One license is required per MWSynthRef board to use the DUT low source.
LowPhaseNoiseDUTRefSource	Low Phase Noise DUT/Clock Reference Source	Counted license. One license is required per MWSynthRef board to use the low phase noise DUT/Clock reference source.
NoiseSource	On-Board Source Noise	Counted license. One license is required per measurement board.
OnBoardDUALAWGSynthDR	On-Board Dual AWG (Synthesizer DR or LO)	Counted license. One license is required per dual AWG board for modulated source.

For more information, see [IG-XL Licensing](#) in the IG-XL Administration section.

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UltraWave 12G Examples

UltraWave 12G Examples

This section contains IG-XL workbook examples showing all IG-XL data and VBT code necessary for a complete running program:

- MWSource-MWReceiver Loopback Test
- MWModulatedSource-MWReceiver AM Signal Test

For more information, see:

- UltraWave 12G Instrument Programming
- Individual MW instruments in the VBT Syntax Reference, such as MWSource

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UltraWave 12G Examples : MWSource-MWReceiver Loopback Test

MWSource-MWReceiver Loopback Test

The MWSource-MWReceiver loopback test verifies the connection from the MWSource to the MWReceiver through a cable and tests the amplitude of a sourced RF signal. The test program tests two sites.

A MWSource generates CW signals on two ports. Two MWReceivers downconvert the two signals and digitize them. (A site loop is not required because there is a receiver and digitizer for each site.) The amplitudes are calculated using a DFT DSP method.

The program runs both online and offline. Offline, a function in the DSP module generates a received waveform for the DFT routine. Since the program is dual site, two cables are required to run online.

The MW Measure board is installed in test head slot 2. A MW Synthesizer LO board must be internally connected to the MW Measure board, but the test head slot does not affect the channel map.

This workbook example contains the following components:

- | | |
|---|-----------------|
| • | DUT Connections |
| • | DataTool Sheets |
| • | VBT Code |
| • | DSP Code |

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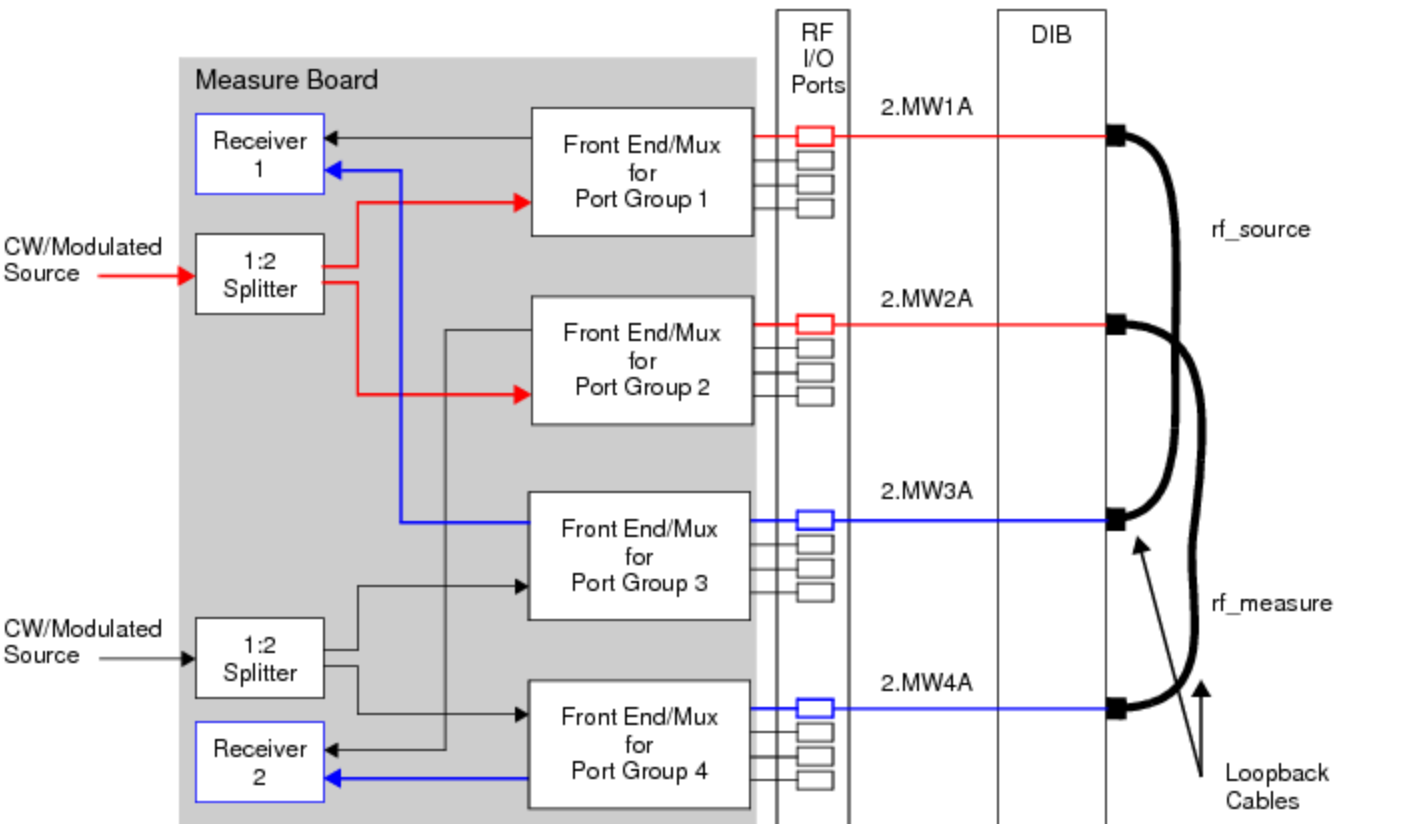
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UltraWave 12G Examples : [MWSource-MWReceiver Loopback Test](#) : DUT Connections

DUT Connections



UltraWave 12G MWSource-MWReceiver Loopback Test Connections

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UltraWave 12G Examples : [MWSource-MWReceiver Loopback Test](#) : DataTool Sheets

DataTool Sheets

This workbook example includes the following DataTool sheets:

- Pin Map Sheet
- Channel Map Sheet
- Flow Table Sheet
- Test Instances Sheet
- Mixed Signal Timing Sheet
- Waveform Definition Sheet

Pin Map Sheet

Pin Map

Group Name	Pin Name	Type
	rf_source	Analog
	rf_measure	Analog

Channel Map Sheet

Channel Map

DIB ID:

View Mode: Signal

Device Under Test		Tester Channel		
Pin Name	Package Pin	Type	Site 0	Site 1
rf_source		MW	5.a1	5.a2
rf_measure		MW	5.b1	5.b2

Flow Table Sheet

Flow Table

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					Flow Domain:													
		Gate			Command				Limits		Datalog Display			Bin Number		Sort Number		
Label	Enable	Job	Part	Env	Opcode	Parameter	TName	TNum	LoLim	HiLim	Scale	Units	Format	Pass	Fail	Pass	Fail	Result
					Test	loopback_cw												
					Use-Limit	loopback_cw	rf_pow	99	-40	20				1	99	1	99	Fail

Test Instances Sheet

Test Instances

Test Procedure			
Test Name	Type	Name	Called As
loopback_cw	VBT	RF_meas_loopback	

The test instance named loopback_cw has the following test-specific parameters and values as shown in the Instance Editor:

Instance Editor [VBT Test 'RF_meas_loopback']

Instance:

Parameters | Timing and Levels | Limits

Name	Value
SrcPin	rf_source
RecvPin	rf_measure
RF_Freq	900000000
RF_power	-20
IF_freq	2512500
capture_FS	12500000
capture_size	1000

Mixed Signal Timing Sheet

The MW instrument does not use the Mixed Signal Timing sheet.

Waveform Definition Sheet

The MW instrument does not require a Waveform Definition sheet.

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UltraWave 12G Examples : [MWSource-MWReceiver Loopback Test](#) : VBT Code

VBT Code

```
Public Function RF_meas_loopback(SrcPin As PinList, RecvPin As PinList, _  
    RF_Freq As Double, RF_power As Double, _  
    IF_freq As Double, capture_FS As Double, _  
    capture_size As Long) As Long
```

```
Dim RfSignal As String
```

```
Dim RfCapturedWave As New DSPWave
```

```
Dim measured_power As New SiteDouble
```

```
On Error GoTo errHandler
```

```
RfSignal = "RfCapture"
```

```
With TheHdw.MWSource(SrcPin)
```

```
    .amplitude = RF_power
```

```
    .Frequency = RF_Freq
```

```
End With
```

```
With TheHdw.MWReceiver(RecvPin)
```

```
    .Signals.Add RfSignal
```

```
    If TheExec.TesterMode = testModeOnline Then
```

```
        RfCapturedWave = .Signals(RfSignal).DSPWave.Pins(RecvPin)
```

```
    End If
```

```
End With
```

```
With TheHdw.MWReceiver(RecvPin)
```

```
    .AmplitudeExpected = RF_power
```

```
.CenterFrequency = RF_Freq
.IFFrequency = IF_freq
.capture.SampleRate = capture_FS
.capture.SampleSize = capture_size
.GainSetup = tlmwGainSetupNormal
End With

TheHdw.MWSource(SrcPin).Connect
With TheHdw.MWReceiver(RecvPin)
    .Connect
    .Signals(RfSignal).Trigger
    .WaitForCaptureDone
    .Disconnect
End With
TheHdw.MWSource(SrcPin).Disconnect

TheHdw.DSP.ExecutionMode = tlDSPModeHostDebug

Dim CurrSite As Variant
If TheExec.TesterMode = testModeOffline Then
    For Each CurrSite In TheExec.Sites
        Call rundsp.create_offline_capture(RF_power, IF_freq, capture_FS, _
            capture_size, RfCapturedWave)
    Next CurrSite
End If

rundsp.measRF_power RfCapturedWave, IF_freq, measured_power
Call TheExec.Flow.TestLimit(measured_power, Unit:=unitCustom, _
    customunit:="dBm", forceresults:=tlForceFlow, ScaleType:=scaleNoScaling)

RF_meas_loopback = TL_SUCCESS

Exit Function
errHandler:
    If AbortTest Then Exit Function Else Resume Next
End Function
```

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UltraWave 12G Examples : [MWSource-MWReceiver Loopback Test](#) : DSP Code

DSP Code

```
Public Function measRF_power(ByVal dsp_in As DSPWave, _
    ByVal IF_freq As Double, _
    ByRef measured_power_dBm As Double) As Long

    Dim bin As Long, phase_ignore As Double
    Dim ampl As Double
    bin = IF_freq / dsp_in.FrequencyResolution
    ampl = dsp_in.CalcAmplitudeFromTimeDomain(bin, phase_ignore)
    measured_power_dBm = (10# / Log(10#)) * Log(1000# * ampl * ampl _
        / (2 * 50#))

    measRF_power = TL_SUCCESS
End Function

Public Function create_offline_capture(ByVal power_dBm As Double, _
    ByVal IF_freq As Double, _
    ByVal capture_rate As Double, _
    ByVal sample_size As Long, _
    ByRef capture As DSPWave) As Long

    Dim omega As Double
    Dim ampl As Double

    ampl = 10# ^ ((power_dBm - 10#) / 20#)
    omega = 2 * Pi * IF_freq / capture_rate
```

```
Call capture.CreateSin(omega, 0, sample_size)
capture = capture.Multiply(ampl)
capture.SampleRate = capture_rate
create_offline_capture = TL_SUCCESS
End Function
```

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UltraWave 12G Examples : MWModulatedSource-MWReceiver AM Signal Test

MWModulatedSource-MWReceiver AM Signal Test

The MWModulatedSource-MWReceiver AM Signal test verifies the connection from the MWModulatedSource to the MWReceiver through a cable and tests the power in the sidebands of an amplitude modulated RF signal. The test program tests two sites.

The MWModulatedSource generates an AM signal using IF signal data. The signal is connected at two ports. Two MWReceivers downconvert the signal and digitize it. (A site loop is not required because there is a receiver and digitizer for each site.) The sideband amplitudes are calculated using DFT DSP methods.

The program runs both online and offline. Offline, a function in the DSP module generates a received waveform for the DFT routine. Since the program is dual site, two cables are required to run online.

The MW Measure board is installed in test head slot 2. A MW Synthesizer LO board must be internally connected to the MW Measure board, but the test head slot does not affect the channel map.

This workbook example contains the following components:

- | | |
|---|-----------------|
| • | DUT Connections |
| • | DataTool Sheets |
| • | VBCode |
| • | DSP Code |

		
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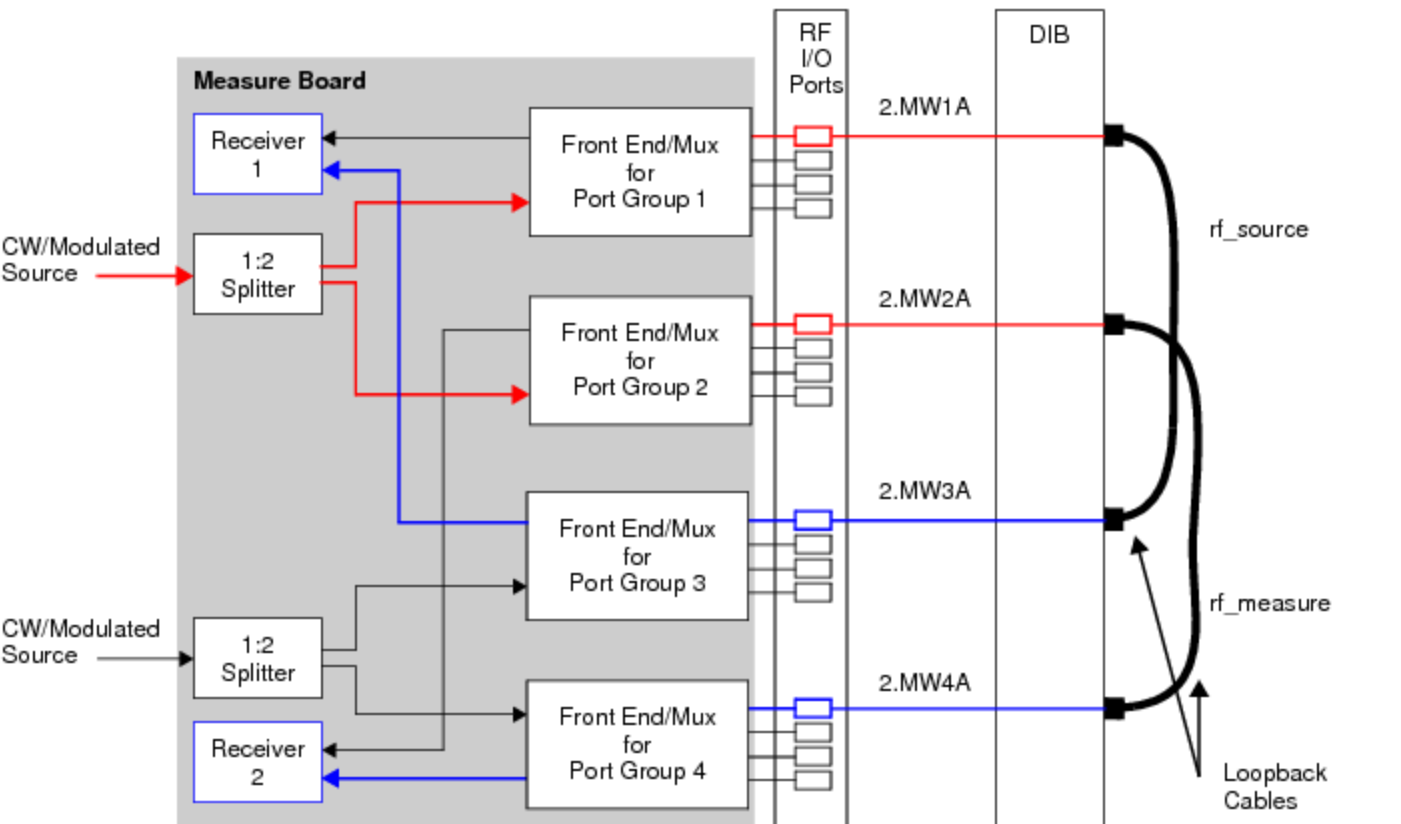
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UltraWave 12G Examples : [MWModulatedSource-MWReceiver AM Signal Test](#) : DUT Connections

DUT Connections



MWModulatedSource-MW Receiver AM Signal Test Connections

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UltraWave 12G Examples : [MWModulatedSource-MWReceiver AM Signal Test](#) : DataTool Sheets

DataTool Sheets

This workbook example includes the following DataTool sheets:

- Pin Map Sheet
- Channel Map Sheet
- Flow Table Sheet
- Test Instances Sheet
- Mixed Signal Timing Sheet
- Waveform Definition Sheet

Pin Map Sheet

Pin Map

Group Name	Pin Name	Type
	rf_source	Analog
	rf_measure	Analog

Channel Map Sheet

Channel Map

DIB ID: View Mode:

Device Under Test		Tester Channel		
Pin Name	Package Pin	Type	Site 0	Site 1
rf_source		MW	2.MW1A	2.MW2A
rf_measure		MW	2.MW3A	2.MW4A

Flow Table Sheet

Flow Table



					Flow Domain:					
					Gate		Command			
Label	Enable	Job	Part	Env	Opcode	Parameter	TName	TNum	LoLim	HiLim
					Test	loopback_am_mod				
					Use-Limit	loopback_am_mod	carrier_pow	100		-20
					Use-Limit	loopback_am_mod	USB_pow	101		-20
					Use-Limit	loopback_am_mod	LSB_pow	102		-20

Test Instances Sheet

Test Instances

Test Procedure		
Test Name	Type	Name
loopback_am_mod	VBT	modsource_am

The test instance named loopback_am_mod has the following parameters and values as shown in the Instance Editor:

Name	Value	Comment
Sourcepin	rf_source	...
Capturepin	rf_measure	...
Sourcepower	-20	...
Sourcefreq	100000000	...
MWRecv_IF_Freq	5000000	...
AM_carrier_IF_Freq	2000000	...
Baseband_Freq	10000	...
modsource_FS	20000000	...
AM_waveform_size	20000	...
AM_Mod_Index	0.5	...
MWRecv_sample_rate	25000000	...
MWRecv_sample_size	25000	...

Mixed Signal Timing Sheet

The MW instrument does not use the Mixed Signal Timing sheet.

Waveform Definition Sheet

The MW instrument does not require a Waveform Definition sheet.

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UltraWave 12G Examples : [MWModulatedSource-MWReceiver AM Signal Test](#) : VBT Code

VBT Code

```
Public Function modsource_am(ByVal Sourcepin As pinlist, _
    ByVal Capturepin As pinlist, _
    ByVal Sourcepower As Double, _
    ByVal Sourcefreq As Double, _
    ByVal MWRecv_IF_Freq As Double, _
    ByVal AM_carrier_IF_Freq As Double, _
    ByVal Baseband_Freq As Double, _
    ByVal modsource_FS As Double, _
    ByVal AM_waveform_size As Double, _
    ByVal AM_Mod_Index As Double, _
    ByVal MWRecv_sample_rate As Double, _
    ByVal MWRecv_sample_size As Double)
```

' Parameter declarations

```
Dim CrestFactor As Double
Dim baseband_signal As New DSPWave
Dim LO_Signal As New DSPWave
Dim modulation_signal As New DSPWave
Dim Site As Variant
Dim AM_capture_wave As New DSPWave
Dim carrier_power As New SiteDouble
Dim USB_power As New SiteDouble
Dim LSB_power As New SiteDouble
```

Dim mwrecv_spectrum_reversed As New SiteBoolean

```
modulation_signal = AM_modulated_IF(AM_carrier_IF_Freq, Baseband_Freq, _
    AM_Mod_Index, modsource_FS, AM_waveform_size, CrestFactor)
```

' Add the Amplitude Modulated signal in DSPwave to the Wave Definitions

' Collection

If Not (TheExec.WaveDefinitions.Exists("AM_Wavedef")) Then

Call TheExec.WaveDefinitions _

.CreateWaveDefinition("AM_Wavedef", modulation_signal, False)

End If

' Set up MWModulatedSource

TheHdw.MWModulatedSource(Sourcepin).Signals.Add "am_signal"

TheHdw.MWModulatedSource(Sourcepin).Signals.DefaultSignal = "am_signal"

With TheHdw.MWModulatedSource(Sourcepin).Signals("am_signal")

.WaveDefinitionName = "AM_Wavedef"

.Type = tlMWSignalTypeIF

.IFFrequency = AM_carrier_IF_Freq ' this frequency is upconverted to the
center frequency

.ReverseSpectrum = True ' AM signal is symmetric, so this does not matter.

.SampleRate = modsource_FS

.SampleSize = AM_waveform_size

.CrestFactor = CrestFactor

.Frequency = Sourcefreq

.amplitude = Sourcepower

.LoadSettingsConnect

.Start tlStartImmediately

End With

' Set up the MWReceiver

With TheHdw.MWReceiver.Pins(Capturepin)

.Signals.Add "am_mod_cap"

.AmplitudeExpected = Sourcepower

.CenterFrequency = Sourcefreq

.IFFrequency = MWRecv_IF_Freq

```

.IFFilter = 10000000#
.Capture.SampleSize.Value = MWRecv_sample_size
.Capture.SampleRate = MWRecv_sample_rate

AM_capture_wave = .Signals("am_mod_cap").DSPWave.Pins(Capturepin)
mwrecv_spectrum_reversed = .IsSpectrumReversed(Sourcefreq)

.Connect
End With

' Trigger the MWReceiver
TheHdw.MWReceiver.Pins(Capturepin).Connect
TheHdw.MWReceiver.Pins(Capturepin).Signals("am_mod_cap").Trigger
TheHdw.MWReceiver.Pins(Capturepin).WaitForCaptureDone

' Retrieve the DSP wave handle, MWReceiverDSPWave

'-----
' For offline simulation only
' Import the saved DSP wave.
If TheExec.TesterMode = testModeOffline Then
    Dim ampl As Double
    ampl = 10# ^ ((Sourcepower + (CrestFactor - 3#) - 10#) / 20#)
    For Each Site In TheExec.Sites
        ' Should simulate spectrum reversal; AM is symmetric, however
        AM_capture_wave = AM_modulated_IF(MWRecv_IF_Freq, Baseband_Freq, _
            AM_Mod_Index, MWRecv_sample_rate, MWRecv_sample_size) _
            .Multiply(ampl)
    Next Site
End If
'-----

' AM_capture_wave(0).Plot "am_capture 0"
' AM_capture_wave(1).plot "am_capture 1"

' Call the DSP routine
Call rundsp.am_sb_power(AM_capture_wave, mwrecv_spectrum_reversed, _
    MWRecv_IF_Freq, Baseband_Freq, carrier_power, USB_power, LSB_power)

```

' Datalog the results

```
Call TheExec.Flow.TestLimit(resultval:=carrier_power, unit:=unitCustom, _
                             customUnit:="dBm", ScaleType:=scaleNoScaling, _
                             forcerevents:=tlForceFlow)
```

```
Call TheExec.Flow.TestLimit(resultval:=USB_power, unit:=unitCustom, _
                             customUnit:="dBm", ScaleType:=scaleNoScaling, _
                             forcerevents:=tlForceFlow)
```

```
Call TheExec.Flow.TestLimit(resultval:=LSB_power, unit:=unitCustom, _
                             customUnit:="dBm", ScaleType:=scaleNoScaling, _
                             forcerevents:=tlForceFlow)
```

' Disconnect the instruments

```
With TheHdw.MWModulatedSource.Pins(Sourcepin)
    .Signals.Stop tlStopImmediately
    .Disconnect
End With
```

```
TheHdw.MWReceiver.Pins(Capturepin).Disconnect
```

End Function

' The main parameters passed into this function are:

```
,
' IFFreq - the Amplitude Modulated signal Carrier IF Frequency.
' modulation_Freq - frequency at which the amplitude is modulated
' (i.e., the AM signal frequency)
' sample_rate - the sample frequency (in Hz) of the modulated segment
' sample_size - the sample size of the modulated segment
' modulation_index - the Modulation Index for the Amplitude Modulated signal
,
```

' The peak amplitude is normalized to remain in the +/-1.0 range

```
,
' The function returns
,
```

```
' crest_factor_dB: ratio of peak to RMS power, in dB
,
```

```
Public Function AM_modulated_IF(ByVal IFFreq As Double, _
    ByVal modulation_freq As Double, _
    ByVal modulation_index As Double, _
    ByVal sample_rate As Double, _
    ByVal sample_length As Long, _
    Optional ByRef crest_factor_dB As Double) _
    As DSPWave
```

```
Dim baseband_signal As New DSPWave
Dim carrier_signal As New DSPWave
Dim modulation_signal As New DSPWave
```

```
' Create AM baseband "information" signal: will modulate the carrier
' amplitude. Determines the sideband spacing of the modulated waveform.
```

```
Call baseband_signal _
    .CreateCos(2 * PI * modulation_freq / sample_rate, 0, sample_length)
baseband_signal.SampleRate = sample_rate
```

```
' Create carrier signal: sets the center frequency of the AM waveform
```

```
Call carrier_signal.CreateCos(2 * PI * IFFreq / sample_rate, 0, sample_length)
carrier_signal.SampleRate = sample_rate
```

```
' Create the AM signal by multiplying the two sine waves together
'  $A[1 + (\text{mod\_index})\cos(\text{baseband\_freq})]\cos(\text{carrier\_freq})$  or
'  $A[\cos(\text{carrier\_freq}) + (\cos(\text{carrier\_freq}) * \cos(\text{baseband\_freq}) * \text{mod\_index})]$ 
```

```
modulation_signal = baseband_signal.Multiply(modulation_index)
modulation_signal = _
    carrier_signal.Multiply(modulation_signal).Add(carrier_signal)
```

```
' Normalize to +/-1.0
Dim peak_ampl As Double
peak_ampl = modulation_signal.CalcMaximumMagnitude
modulation_signal = modulation_signal.Multiply(1# / peak_ampl)
```

```
modulation_signal.SampleRate = sample_rate
```

```
' Calculate the CrestFactor of the AM signal
crest_factor_dB = 20 * Log(modulation_signal.CalcMaxMagnitude / _
    modulation_signal.CalcRMS) / Log(10#)
Set AM_modulated_IF = modulation_signal
```

End Function

DSP Code

```
Public Function am_sb_power(ByVal Capture As DSPWave, _
    ByVal spectrum_reversed As Boolean, _
    ByVal IFFreq As Double, _
    ByVal Freq_BB As Double, _
    ByRef carrier_dBm As Double, _
    ByRef power_dBm_hi As Double, _
    ByRef power_dBm_lo As Double) As Long
```

```
' Parameter Declarations
```

```
Dim bin_hi As Long, bin_lo As Long, bin_carrier As Long
Dim Fres As Double, ampl_hi As Double, ampl_lo As Double
Dim ampl_carrier As Double
Dim bin_spacing As Long
```

```
' Calculate Frequency resolution of the capture
```

```
Fres = Capture.FrequencyResolution
bin_carrier = IFFreq / Fres
bin_spacing = Freq_BB / Fres
```

```
' Calculate the bin separation of the base band signal
```

```
' Calculate the bin number of the upper and lower side band
```

```
If spectrum_reversed Then
    bin_hi = bin_carrier - bin_spacing
    bin_lo = bin_carrier + bin_spacing
Else
    bin_hi = bin_carrier + bin_spacing
    bin_lo = bin_carrier - bin_spacing
End If
```

' Calculate the Amplitudes of the LSB and USB.

```
ampl_carrier = Capture.CalcAmplitudeFromTimeDomain(bin_carrier)
ampl_hi = Capture.CalcAmplitudeFromTimeDomain(bin_hi)
ampl_lo = Capture.CalcAmplitudeFromTimeDomain(bin_lo)
```

' Convert the Amplitudes calculated above to power.

```
carrier_dBm = (10# / Log(10#)) * _
                Log(1000# * ampl_carrier * ampl_carrier / (2 * 50))
power_dBm_hi = (10# / Log(10#)) * Log(1000# * ampl_hi * ampl_hi / (2 * 50))
power_dBm_lo = (10# / Log(10#)) * Log(1000# * ampl_lo * ampl_lo / (2 * 50))
```

```
am_sb_power = 0 ' TL_SUCCESS
```

End Function

UW12GSubsys_Cal.3.1

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UltraWave 12G Instrument Calibration

UltraWave 12G Instrument Calibration

The MW instruments require calibration to meet performance specifications. Instrument specifications depend on all three calibration levels supported by the UltraFLEX test system:

Level 1	Internal calibration (also called automatic calibration or autocal)
Level 2	Maintenance programs
Level 3	Internal reference calibration to traceable external standards

Most MW Level 3 calibrations are performed at the factory before shipment. Level 1 and 2 calibrations are performed in the field, in the environment where the tester is used. In addition to the three levels of calibration, the MWSource and MWReceiver instruments perform calibrations at run time. Run-time calibrations include autocal and program-specific levelling. The three calibration levels and the UltraFLEX calibration strategy are described in [Levels of Calibration](#) and in [Maintenance Environment Help](#). This section includes the following topic:

- [Run-Time Calibration](#)

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UW12GSubsys_Cal.3.2

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UltraWave 12G Instrument Calibration : Run-Time Calibration

Run-Time Calibration

The MWSource and MWReceiver meet specifications by automatically performing calibrations at run time. These run-time calibrations are:

- | | |
|---|-------------------------------------|
| • | Source Level Calibration (Leveling) |
| • | Receiver Gain Calibration |

Source Level Calibration (Leveling)

Source level calibration, also called “leveling,” finely calibrates the MW sources (MWSource, MWModulatedSource, and MWMultiToneSource) at each frequency set in a program. Leveling takes place at run time.

Leveling recalibrates the reference power level for a frequency and channel using the internal port power sensor and the factory calibration data. The source then uses the reference cal data and other calibration data to determine source settings for all other levels at that frequency.

Leveling occurs the first time a specific frequency is set for the MWSource or any other MW source type. Later VBT statements setting the same frequency use cached data and do not level, unless leveling is invoked using the Pins.LevelImmediate VBT property for the source type. For example, see the LevelImmediate property for the Pins interface of TheHdw.MWSource. Leveling can also be invoked for all frequencies used in a test program using the Pins.LevelRefresh VBT method for a source.

Receiver Gain Calibration

Receiver gain calibration, or system gain calibration, finely calibrates the MWReceiver gain for a specific instrument setup at one input frequency, at run time.

Receiver gain calibration connects the receiver to the source through an internal loopback path and transfers an absolute reference level determined during factory cal to the receiver. The gain for the path is calibrated to the receiver input. Receiver gain is measured for each combination of parameter settings for full instrument accuracy.

Only the multiplexer that includes the channel and port for the assigned pin is calibrated.

As with source leveling, receiver gain calibration occurs the first time a specific instrument setup is used for the MWReceiver. Later VBT statements with the same setup use cached data and do not run calibration. Gain calibration can also be invoked for all setups used in a test program using the Pins.LevelRefresh VBT method for the receiver.

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UW12GSubsys_Displays.5.1

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UltraWave 12G Displays

UltraWave 12G Displays

The UltraWave 12G subsystem supports several displays for test programming and debugging:

Microwave Board Debug Display	This display provides physical board-level views of the Microwave instrument hardware (boards) installed and configured in the tester. It shows the boards, how they are connected, and how the hardware functions are used by logical instruments. The Microwave Board Debug Display is available in TDE.
Microwave Debug Display	This display shows the state of one or more MW logical instruments in the context of a site. Each debug display features the control and display functions of a logical instrument, but hides the physical hardware. The Microwave Debug Display is available in TDE.
Smith Display	The Smith Display plots s-parameter data on a Smith Chart, mapping coefficients of reflection to impedance. You create s-parameter data with the MWVNA logical instrument. It is available from the Windows command line. See Smith Display.

This section describes features specific to the UltraWave 12G Microwave displays. For more information on using debug displays, see the following:

- UltraWave12G Microwave Board Debug Display
- UltraWave 12G Microwave Debug Display

- [Using Debug Displays in TDE](#)
-
- [Producing Code from Debug Displays](#)

IG-XL also includes the Vector Scope, a display tool that plots complex data. This display is useful for analyzing modulated RF signal waveforms. Vector Scope is available from either the Windows command line or by using the DSPWave Plot method in IG-XL debug mode. For more information, see [Vector Scope](#).

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UW12GSubsys_Displays.5.2



UltraWave 12G Displays : UltraWave12G Microwave Board Debug Display

UltraWave12G Microwave Board Debug Display

The Microwave Board Debug Display for the UltraWave 12G subsystem provides two kinds of board level views of the instrument hardware in a tester. The different views are displayed, one at a time, in the display window as tabs, one tab for each view:

Navigator	This tab shows the Synthesizer and Measure boards in the tester configuration as blocks with internal tester connection terminals. You can read and display connectivity information, launch board views, and navigate to the different board views. The display can only include one Navigator tab.
Board view	This tab shows a high-level diagram of a board with a high-level schematic diagram of the main board functions with signal routing and settings. The display can include more than one board view, each with a tab. You can read and display the primary settings, basic signal paths, and connections for logical instruments. A board view has links to the logical instrument debug displays. The Measure board tab is useful for examining interactions between logical instrument paths. For example, you can view and compare the signal paths for a MWModulatedSource and a MWReceiver.

The board display shows only the hardware in your tester configuration. Displays are updated to reflect the current logical instrument connections and settings. In some cases, you can change instrument settings from board displays.

For more information, see the following topics:

- [Starting the UltraWave 12G Microwave Board Debug Display](#)
- [Navigator Tab and Controls](#)
- [Board Tabs and Controls](#)

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UltraWave 12G Displays : [UltraWave12G Microwave Board Debug Display](#) : Starting the UltraWave 12G Microwave Board Debug Display

Starting the UltraWave 12G Microwave Board Debug Display
Before you can start the [Microwave Board Debug Display](#), the active program must:

- | | |
|---|--|
| • | Be in debug mode |
| • | Use at least one Microwave pin |
| • | Be validated |

To launch the [Microwave Board Debug Display](#), on the [IG-XL Tools](#) tab, in the Launch group, select Subsystems>Microwave Board. The Microwave Board Debug Display appears with the Navigator tab shown. See [Navigator Tab and Controls](#).

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UW12GSubsys_Displays.5.4

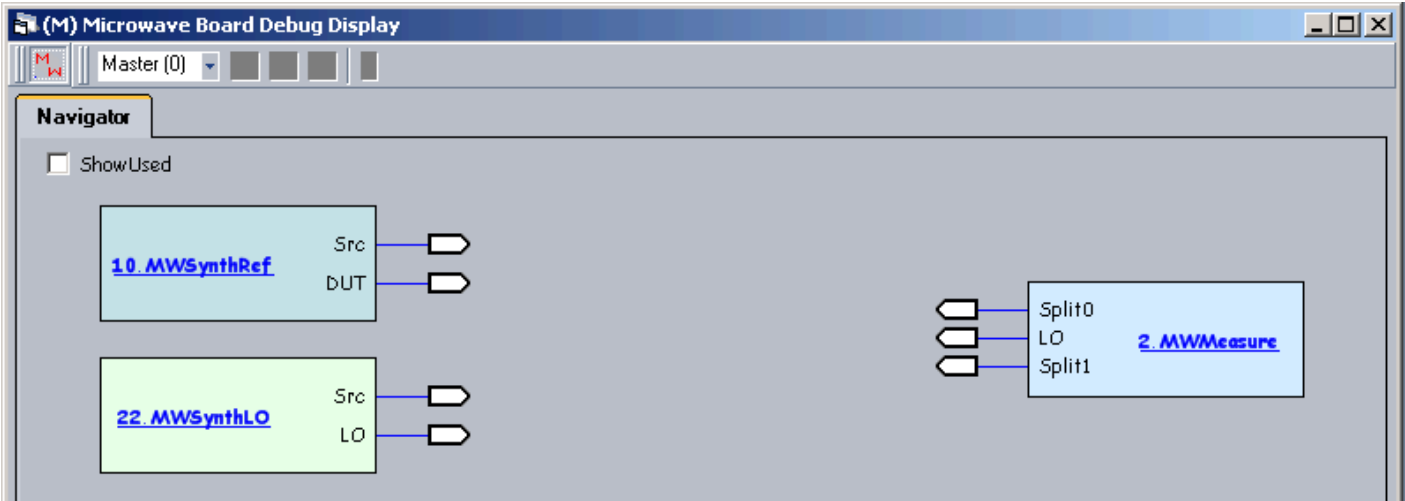
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UltraWave 12G Displays : UltraWave12G Microwave Board Debug Display : Navigator Tab and Controls

Navigator Tab and Controls

When it opens, the Microwave Board Debug Display shows the Navigator tab. The example in the figure shows a configuration with two Synthesizer boards and one Measure board.



Board Debug Display Navigator Tab

The display shows all boards in the current tester configuration arranged in two columns as follows:

- Synthesizer boards on the left side
- Measure boards on the right side
- Both board columns in ascending order top to bottom, by slot number

For each board, the Navigator tab shows:

	Slot number
	Board type
	Internal port names and terminals

The Navigator tab has the following controls:

board block	Represents a board. The tooltip shows a list of all connected boards.
board block label	Identifies the slot number and UltraWave 12G board type. Also, as an active link (underlined), the board label does one of the following: - Adds a new board view with tab, if there is not a board tab for the board in that slot. - Selects the tab for the board, if there is already a board tab for the board in that slot.
terminal symbol	Shows internal cable connections between Synthesizer boards and Measure boards. Terminals are active. Click to show connections between a Synthesizer board output and all the inputs of connected Measure boards. The display clears any existing connections before showing new ones. The DUT output of a Reference Synthesizer (MWSynthRef) provides external signals to the DUT; external connections are not shown.
Show Used	Displays only boards used in the current channel map, when checked. Otherwise, the display shows all boards in the current tester configuration.

	
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UW12GSubsys_Displays.5.5



UltraWave 12G Displays : UltraWave12G Microwave Board Debug Display : Board Tabs and Controls

Board Tabs and Controls

When the Microwave Board Debug Display first opens, it shows only the Navigator tab.

To create a new tab with a board view from the Navigator tab:

- Click the label link (underlined) on the block for the board. For example, 2.MWMeasure is the label for a Measure board in slot 2.

A new tab appears with the board label, such as 2-MWMeasure, and the high-level schematic for the board type appears.

If a tab for the board already exists, when you click the label link in the block, the display changes to the tab for the labeled board.

To view an existing board tab from the Navigator tab, do either of the following:

- Click the label link (underlined) on the block for the board. For example, 2.MWMeasure for a Measure board in slot 2.

The display changes to the tab for the labeled board.

- Click the tab for the board.

The display changes to the tab for the board.

To close a board tab:

- Click the red "X" button at the far right on the tab bar, while the tab is selected.

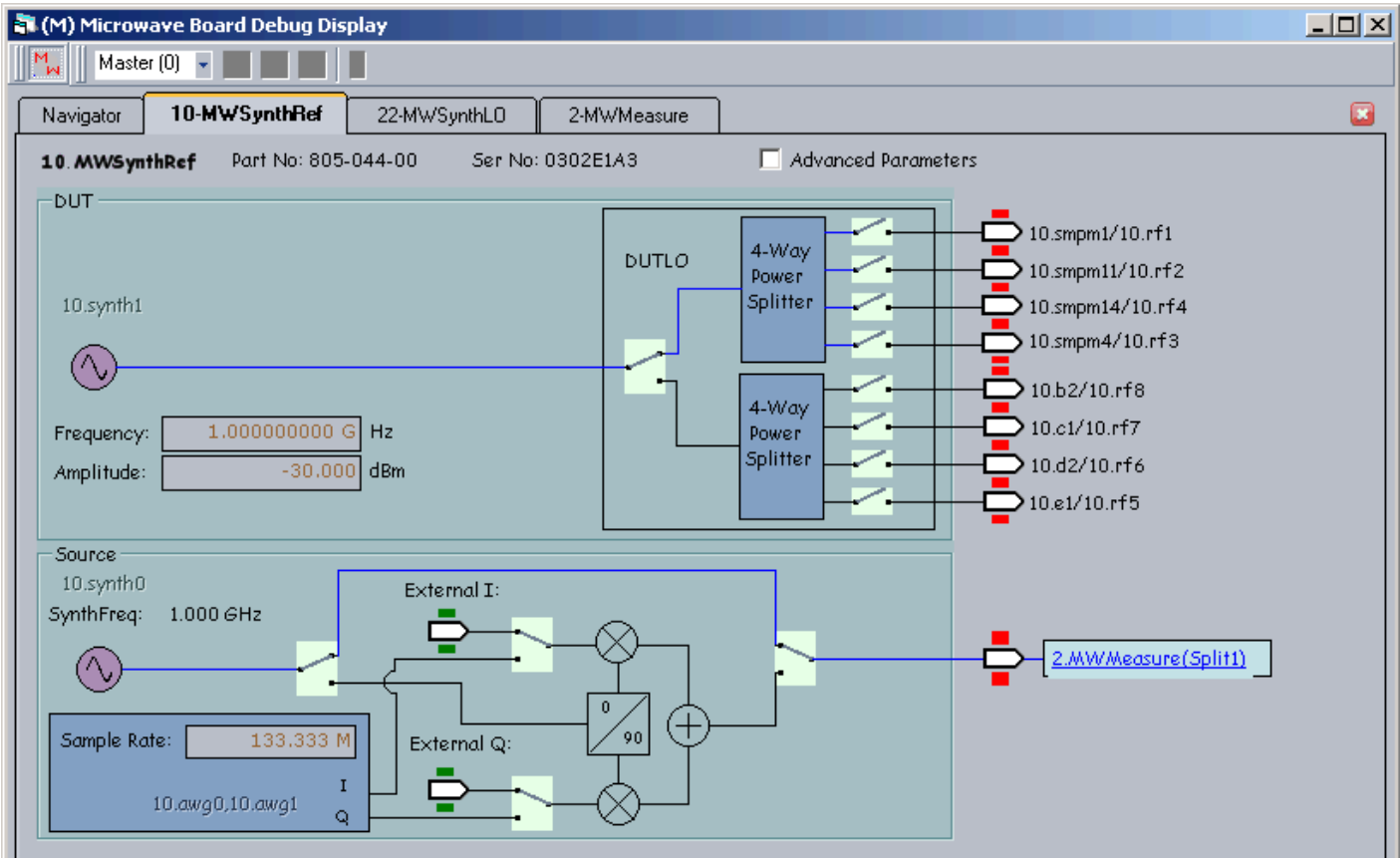
The tab disappears. You can close only board tabs, not the Navigator tab.

The information displayed on a board tab and the controls are specific to the board type:

- [Reference Synthesizer Board Tab Display and Controls](#)
- [LO Synthesizer Board Tab Display and Controls](#)
- [Measure Board Tab Display and Controls](#)

Reference Synthesizer Board Tab Display and Controls

When you create or select a Reference Synthesizer board tab, the display appears:



Reference Synthesizer Board Tab

The display shows both synthesizers and sources on a Reference Synthesizer board, along with the primary settings, basic signal paths, and connections.

The DUT Reference shows output terminals at the DIB interface (RF ports or pins). The CW/modulated source shows the internal output connection to a splitter input on one or more Measure boards. Up to four Measure boards can be connected and shown.

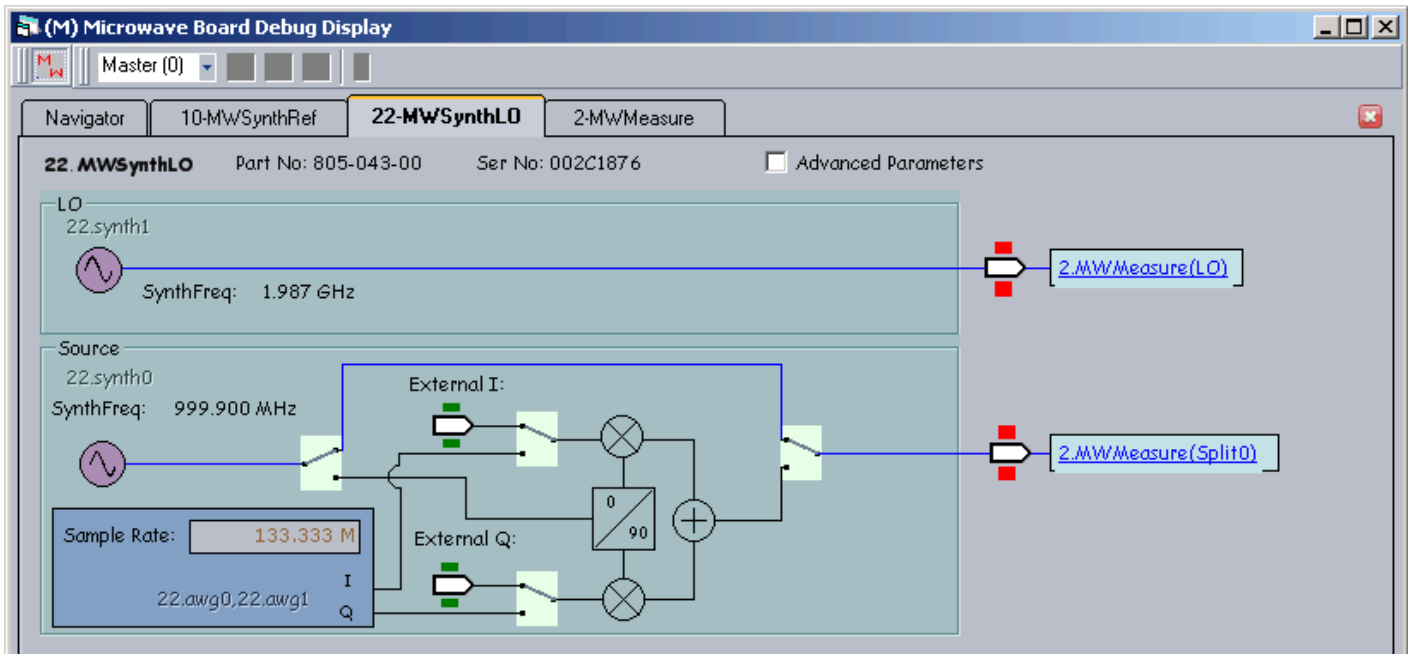
The Reference Synthesizer Board Tab has the following controls:

Frequency	Sets and displays the sourced signal frequency for either the DUT LO signals at one to four RF DIB ports, or the DUT Clock signals at one to
-----------	--

	four DIB pins. The tooltip for this field shows the allowable frequency range.
Amplitude	Sets and displays the sourced signal amplitude for either the DUT LO signals at one to four RF DIB ports, or the DUT Clock signals at one to four DIB pins. The tooltip for this field shows the allowable amplitude range.
DUT LO or DUT Clock output terminal	Tooltip shows pin name and logical instrument name, only if instrument is connected.
SynthFreq	Displays the synthesizer frequency for the CW/modulated source.
source output terminal	Displays names of connected Measure boards. Also, as an active link (underlined), the board label selects the tab for the measure board.
Sample Rate	Displays the sample rate of the internal I and Q modulation signals.
relays	Displays the connection state for: • DUT synthesizer set to either DUT LO or DUT Clock • Individual DUT LO and DUT Clock outputs connected or disconnected • Source set to either CW or modulated • I and Q modulation signals sources set to either external or internal
Advanced Parameters	Displays detailed information about the internal states of some MW hardware settings, when checked. This information is only useful during support from Teradyne.

LO Synthesizer Board Tab Display and Controls

When you create or select an LO Synthesizer board tab, the display appears:



LO Synthesizer Board Tab

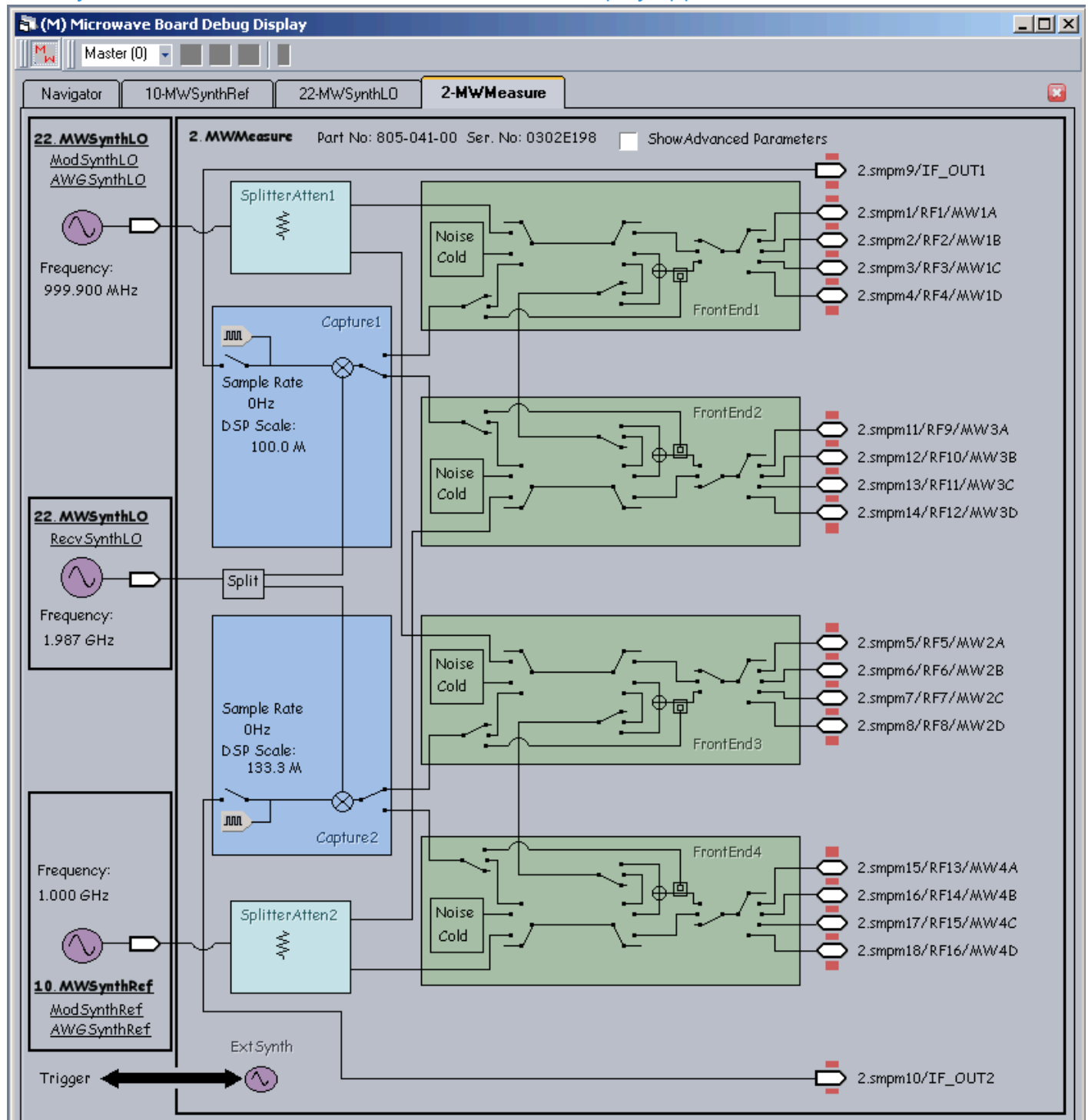
The display shows both synthesizers and sources on an LO Synthesizer board, along with the primary settings, basic signal paths, and connections.

The LO Reference shows the internal output connection to the receiver LO input on one or more Measure boards. The CW/modulated source shows the internal output connection to a splitter input on one or more Measure boards. Up to four Measure boards can be connected and shown. The LO Synthesizer Board Tab has the following controls:

SynthFreq	Displays the synthesizer frequency for the LO synthesizer and the CW/modulated source.
source output terminals	Displays names of the connected Measure boards. Also, as an active link (underlined), the board label selects the tab for the measure board.
Sample Rate	Displays the sample rate of the internal I and Q modulation signals.
relays	Displays the connection state for: - Source set to either CW or modulated. - I and Q modulation signals sources set to either external or internal.
Advanced Parameters	Displays detailed information about the internal states of some MW hardware settings, when checked. This information is only useful during support from Teradyne.

Measure Board Tab Display and Controls

When you create or select a Measure board tab, the display appears:



Measure Board Tab with No Logical Instruments Connected

The display shows the four front ends (MW channels), two receivers with digitizers, and the two splitter/attenuators for a Measure board. In addition, the display shows the synthesizers providing source and LO signals to the Measure board. As with the Synthesizer boards, the Measure board display shows the primary settings, basic signal paths, and connections.

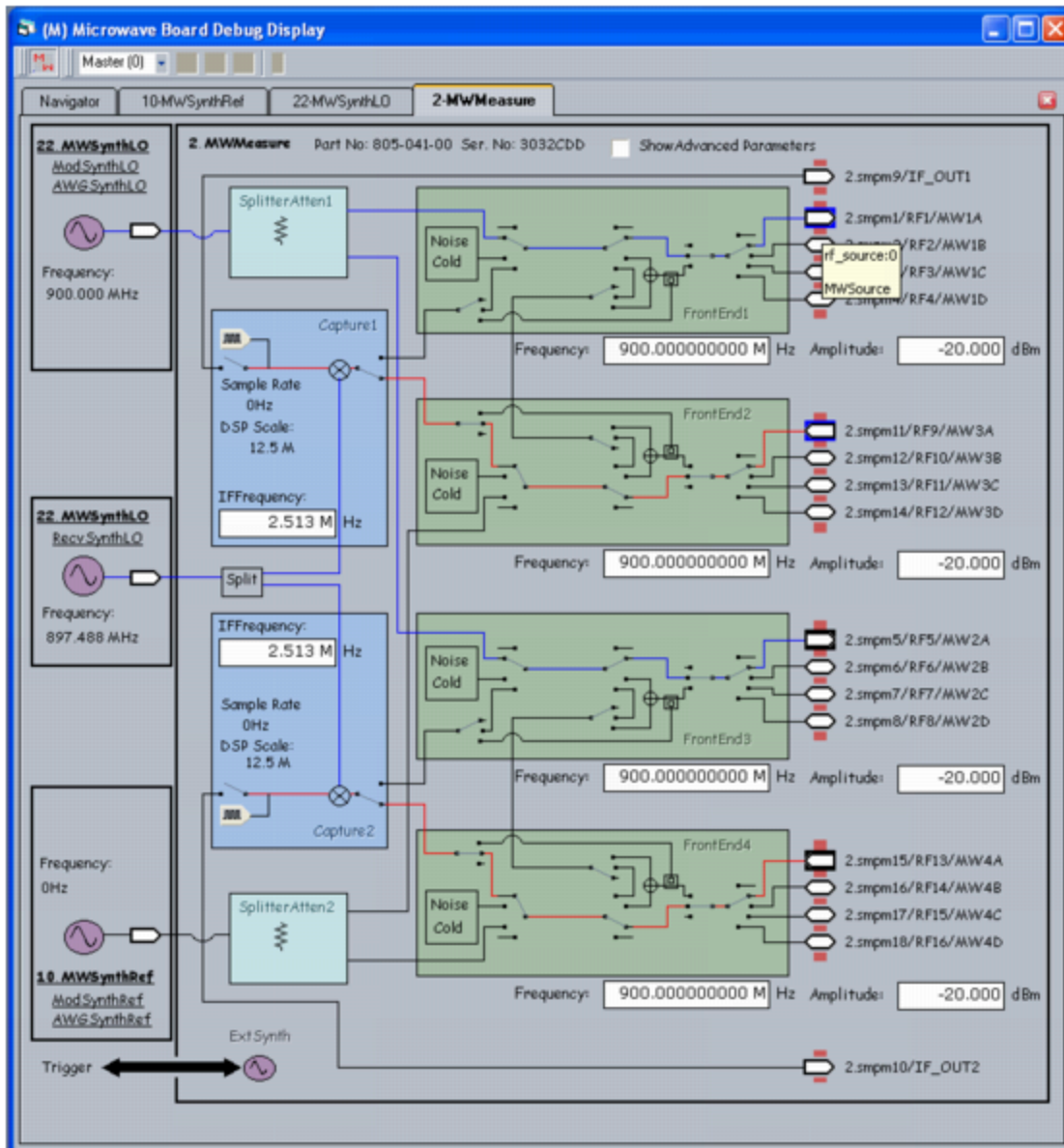
The Measure board tab shows bidirectional terminals representing the multiplexer RF ports at the DIB interface, four ports for each front end. The tab also shows the IF output port for each capture

block (receiver/digitizer).

Note:

The Trigger and ExtSynth in the lower left corner of the display and the bidirectional connection between them are not supported.

When instruments are connected, the display shows more information as shown in the next figure.



Measure Board Tab with Logical Instruments Connected to DUT

In this example, the MWSource and the MWReceiver are connected, for two sites in parallel. In this case, with the site set to the master site (site 0), the Measure board tab shows the following primary settings, signal paths, and DIB connections for connected ports:

- Frequency and amplitude fields for the MWSources under the front-end modules for the connected ports.

- Frequency and amplitude fields for the MWReceivers under the front-end modules for the connected ports.
- IF frequency fields in the capture blocks for the MWReceivers.
- Signal paths through the front-end blocks and into the capture blocks using switch positions, colored signal wires, and an amplifier symbol shown only when LNA is used.
- Connected RF ports using highlighting: a blue background block behind the terminal symbol or symbols for the selected site, and a black background block behind the terminal symbol or symbols for any other sites.
- Connected RF port signal direction by changing from bidirectional input/output symbol (points both right and left) to:
 - Output symbol (points right) for source direction for the MWModulatedSource
 - Input symbol (points left) for receive direction for the MWReceiver
- Connected pin name and logical instrument name for each connected RF port terminal though a tooltip (hover pointer over symbol).

A Measure Board tab has the following controls:

Frequency	Sets and displays the frequency for a source instrument (MWSource, MWMultiToneSource, and MWModulatedSource) and MWReceiver instrument when an assigned pin is connected. The tooltip for this field shows the allowable frequency range. In a synthesizer board block, displays the frequency setting of the synthesizer in the block.
Amplitude	Sets and displays the sourced signal frequency for either the DUT LO signals at one to four RF DIB ports, or the DUT Clock signals at one to four DIB pins. The tooltip for this field shows the allowable amplitude range.

DIB port input or output terminal	Indicates the connection state and port direction for a connected pin, and selects the logical instrument Debug Display for the selected site. For a connected pin: • Terminal symbol is highlighted (blue background block for the selected site or black background block for any other site). • Tooltip shows pin name and logical instrument name. • Click a terminal for the selected site (blue background) to start the Microwave Debug Display and display the logical instrument for the connected pin.
board block label	Identifies a synthesizer slot number and synthesizer board type. Also, as an active link (underlined), the synthesizer board label does either of the following: • Adds a new board view with tab, if there is not a board tab for the board in that slot. • Selects the tab for the board, if there is already a board tab for the board in that slot.
ENRFreq	For the MWNoiseSource only, sets and displays the target ENR frequency in the front-end block.
TargetENR	For the MWNoiseSource only, sets and displays the target ENR in the front-end block.
IFFrequency	For the MWReceiver only, sets and displays the IF frequency in the capture function block.
CrestFactor	For the MWModulatedSource only, sets and displays the crest factor in the Synthesizer board block.
Show Advanced Parameters	Displays detailed information about the internal states of some MW instrument signal paths, when checked. This information is only useful during support from Teradyne.

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UW12GSubsys_Displays.5.6

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UltraWave 12G Displays : UltraWave 12G Microwave Debug Display

UltraWave 12G Microwave Debug Display

The Microwave Debug Display is an interactive tool that shows and controls the state of one or more MW logical instruments in the context of a site. This multi-instrument display overlays the physical hardware of the MW subsystem, featuring the programmable and read-only parameters of the logical instruments.

You can add or remove instruments from the display, as well as expand or collapse instruments on the display to save screen space.

The Microwave Board Debug Display can show instrument settings and resource constraints when you use it with the Microwave Debug Display.

The following topics describe the display controls and how to use the display:

- Starting the UltraWave 12G Microwave Debug Display
- UltraWave 12G Microwave Debug Display Controls
- UltraWave 12G Logical Instrument Display Controls

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UltraWave 12G Displays : UltraWave 12G Microwave Debug Display : Starting the UltraWave 12G Microwave Debug Display

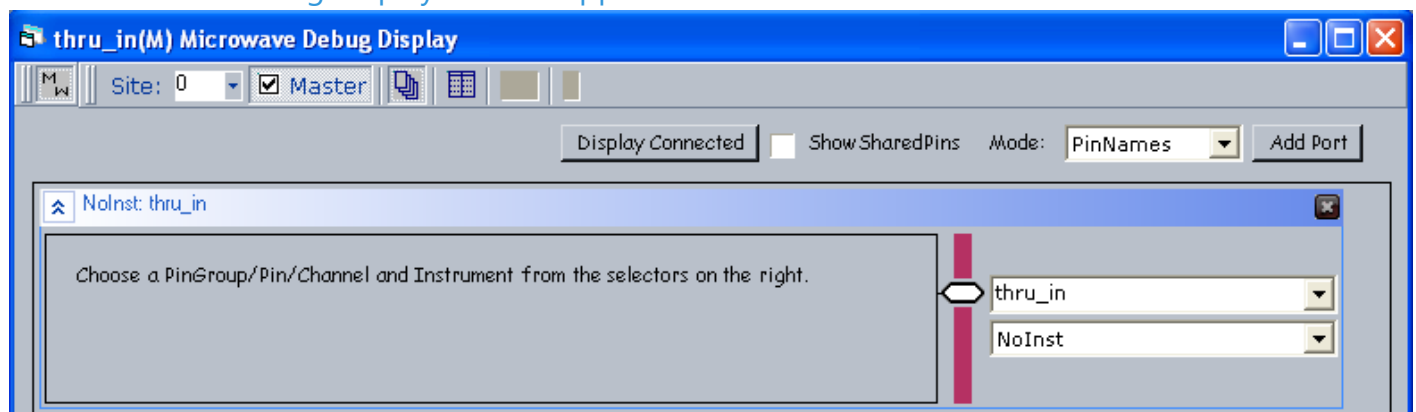
Starting the UltraWave 12G Microwave Debug Display

Before you can start the Microwave Debug Display, the active program must:

- Be in debug mode
- Use at least one Microwave pin, type MW or MWSource
- Be validated

To launch the Microwave Debug Display, on the IG-XL Tools tab, in the Launch group, select Debug Displays>Microwave.

The Microwave Debug Display window appears.



Microwave Debug Display Startup with No Instruments Connected.

At startup, the Microwave debug display shows the instruments currently connected by the test program. If nothing is connected, the display appears showing an empty instrument port (block

and terminal symbol) and prompts you to choose a pin and instrument from the (unlabeled) pin and instrument type drop-down lists just to the right side of the port symbol.

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UltraWave 12G Displays : UltraWave 12G Microwave Debug Display : UltraWave 12G Microwave Debug Display Controls

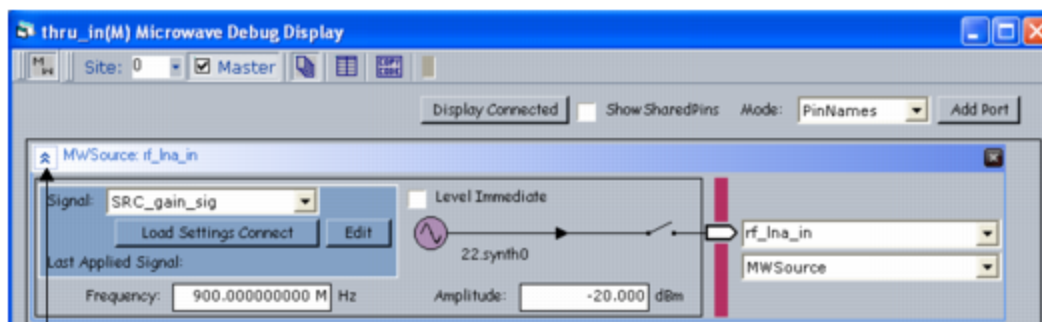
UltraWave 12G Microwave Debug Display Controls
The Microwave Debug Display contains three categories of controls:

- TDE Controls
- General Controls
- Logical Instrument Controls

TDE Controls
The TDE Site selector control and the Apply All Sites, Summary Display, and Copy Generated Code buttons all appear on the Microwave Debug Display toolbar. These controls have the same functions for all debug displays. See Using Debug Displays in TDE.
General Controls
Use the general Microwave Debug Display controls to:

- Add logical instrument ports to the debug display.
- Remove logical instrument ports from the debug display (except for the last port: one must be displayed).
- Expand or collapse an instrument port.

- Set the display mode to pin names or channel names.
- Assign DUT pins or MW channels to a specific instrument type.
- Display all connected instruments in a standard order.
- Show shared pins or shared channels in the instrument port for all connected instruments.
- Identify and locate shared resource conflicts.



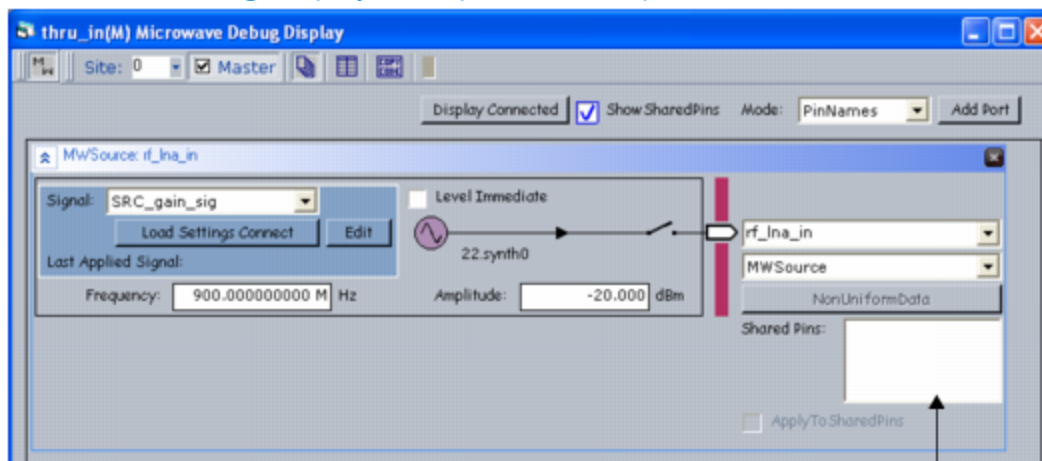
Collapse Icon

Microwave Debug Display, Expanded, with Collapse Icon



Expand Icon

Microwave Debug Display, Collapsed, with Expand Icon



Shared Pins Field

Microwave Debug Display, Expanded, with Shared Pins Field

The source in the example windows above does not use a Signal to control amplitude and frequency.

Logical instrument controls appear on a logical instrument port, which is connected through a switch to a MW port. You use logical instrument controls to read and write logical instrument parameters and connections.

The following general controls are on the UltraWave 12G Microwave Debug Display.

title bar	Displays the selected logical instrument name and pin name.
Display Connected	Displays and sorts the instrument ports, as follows: • Sorts and displays the ports for connected instruments. Ports are shown expanded at the top of the display with sources first, followed by receivers, and then the DUT references. Ports with the same instrument type are sorted by pin groups, then pins, then channels. • Adds a port for each instrument connected in VBT code, if it doesn't have an existing port display. These ports are sorted as listed as above. • Displays ports for unconnected instruments at the bottom of the display. Unconnected ports are collapsed and unsorted.
ShowSharedPins ShowSharedChans	Displays the labeled shared pins field or shared channels field for each instrument port, when checked. Also displays the NonUniformData button and the ApplyToSharedPins or ApplyToSharedChannels check box. (The check box is disabled when there are no shared pins.) The field displays pins for connected ports that share hardware resources.
NonUniformData	Starts an non Uniform Data Summary window and displays nonuniform data for a connected port, only if data is available.

ApplyToSharedPins	Applies changed settings for a port to all pins or channels that share the same hardware resources.
ApplyToSharedChannels	Only displayed when ShowSharedPins or ShowSharedChans is checked.
Mode	Sets the type of data listed on the Pins drop-down list for each logical instrument port displayed. Select from the drop-down list: • PinNames: Lists pin names for all channels mapped to a pin and channel names for all channels mapped to a pin. Pin group names are listed last, in bold. • Channels: Lists channel names for all channels.
Add Port	Adds a logical instrument block and port to the display. The port shows the first pin on the pin name drop-down list. The instrument block remains generic until you select an instrument type from the instrument type drop-down list. A scrollbar appears if there are more ports than the window can display. The maximum number of ports is 128.
pin or channel name	Sets and displays the DUT pin name, pin group name, or the MW channel name associated with the instrument port. Names on the drop-down list depend on the Mode control setting. If Mode is set to PinNames, the list displays pin group names at the end of the list, in bold.
instrument type	Sets and displays the logical instrument type assigned to a port. Select from the drop-down list: • NoInst: No logical instrument is selected. • MWModulatedSource • MWMultiToneSource • MWSource

- [MWNoiseSource](#)
- [MWReceiver](#)
- [MWVNA](#)

When you select a logical instrument, the debug display updates the controls to reflect the state of the instrument.

When a pin is defined in the channel map with tester channel type [MWSource](#) for MW Reference DUT resources and selected in the pin list, only [MWSource](#) appears in the instrument list.

For more information, see [UltraWave 12G Logical Instrument Display Controls](#).

DIB terminal symbol

Indicates the port direction for the selected instrument type. The port direction can be input (points to left), output (points to right), or bidirectional (points to both left and right).

The port direction can change depending on the instrument type selected.

expand/collapse

Expands or collapses the logical instrument display. Click the title bar to collapse or expand the logical instrument port display. The icon changes to indicate the click action.

Expanding or collapsing an instrument block does not change the instrument connections.

remove port button (small x on title bar)

Removes a logical instrument block and port from the display.

Since at least one instrument block must remain on the Microwave Debug Display at all times, the remove button is inactive when only one block is displayed.

This button does not change the instrument connections.

resource conflict

Displays a bar-through-circle ("not allowed") symbol over a connect relay when you connect an instrument. The symbol indicates that you cannot connect the instrument because it will conflict with another instrument.

The relay is disabled. You must resolve the conflict by disconnecting the instrument with the blocking shared resource before you can connect the instrument.

If you point at the relay, a Resource Conflict balloon opens and stays open until you close it.

For the active site, the balloon displays a link to the conflicting pin or channel. The link navigates to the conflicting instrument port. The display adds a port for the conflicting instrument if one is not already open.

Displays resource conflicts with for other sites if ApplyAllSites is set, but without a link to the conflicting pin.

Logical Instrument Controls

Each instrument block displays specific information for the instrument type selected. For a list of logical instrument types, see UltraWave 12G Logical Instrument Display Controls.

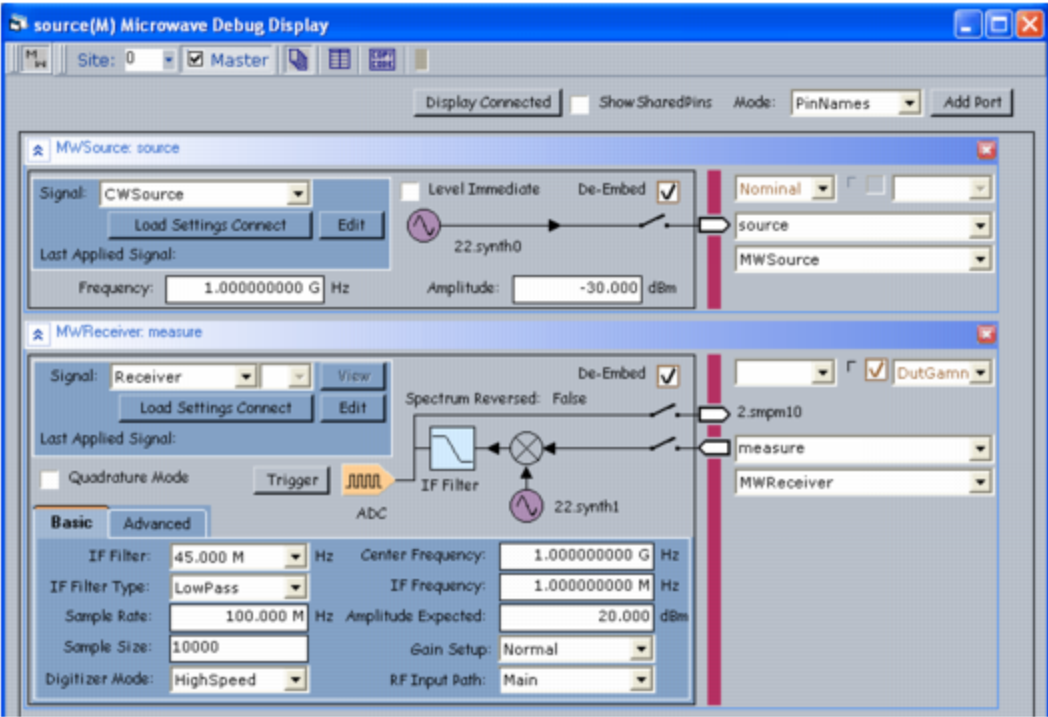
MWDIB Controls for Deembedding

Test programs can include test fixture (DIB) s-parameter data and DUT Gamma information for deembedding one-port and two-port s-parameter measurements. This information is contained in the MWDIB VBT object. If you are using MWDIB in your test program, additional controls in the displays appear. MWDIB controls can be in any of the following states:

- If the test program does not contain any MWDIB syntax, MWDIB controls are hidden in the displays.
- If the MWDIB is used in the test program, but the specified pin has no data, MWDIB information is grayed out in the displays.
- If the MWDIB is used in the test program and the specified pin has data, then MWDIB information is displayed and is interactive.

The following figure shows UltraWave 12G Source and Receiver with MWDIB controls showing Deembed and DUT Gamma information.

For more information, see MWDIB in the VBT Syntax Reference.



Microwave Displays Showing Deembed Controls

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UltraWave 12G Displays : [UltraWave 12G Microwave Debug Display](#) : UltraWave 12G Logical Instrument Display Controls

UltraWave 12G Logical Instrument Display Controls

Each Microwave logical instrument block shows a high-level schematic representation of the instrument. It also provides controls which allow you to set the parameters and perform the operations provided by the Pins VBT language for that instrument.

Each block represents a single logical instrument that is connected to a pin or channel. For descriptions of port display controls for the MW12G logical instruments, see the following sections:

- | | |
|---|---|
| • | UltraWave 12G MWModulatedSource Debug Display |
| • | UltraWave 12G MWMultiToneSource Debug Display |
| • | UltraWave 12G MWNoiseSource Debug Display |
| • | UltraWave 12G MWReceiver Debug Display |
| • | UltraWave 12G MWSource Debug Display |
| • | UltraWave 12G VNA Debug Displays |

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UltraWave 12G Instrument Programming

UltraWave 12G Instrument Programming

At the instrument code level, all MW instruments are programmed in the same general way. As with other kinds of UltraFLEX instruments, you use the VBT language and debug display for each individual MW logical instrument to program the functions of the MW subsystem. In addition, you must connect MW tests to the structure of an IG-XL test program.

When you program several logical instruments, IG-XL checks for resource conflicts at run time. Most conflicts are due to hardware resource mapping and resource sharing. See Detailed Architecture and Resource Mapping and Shared Hardware Functions.

UltraWave 12G instruments do not interact by taking resources currently in use by another instrument. IG-XL run-time errors notify you of resource conflicts and require you to resolve them. This following topics are available:

- [Simulated Configuration Files](#)
- [IG-XL Programming for UltraWave 12G Instruments](#)
- [Using UltraWave 12G in Concurrent Testing](#)
- [UltraWave 12G Pattern Control](#)

The following sections describe how to program each MW logical instrument:

- [UltraWave 12G MWModulatedSource Programming](#)

- [UltraWave 12G MWMultiToneSource Programming](#)
- [UltraWave 12G MWNoiseSource Programming](#)
- [UltraWave 12G MWReceiver Programming](#)
- [UltraWave 12G MWSource Programming](#)
- [UltraWave 12G VNA Programming](#)

For sample MW code, see [UltraWave 12G Examples](#).

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UltraWave 12G Instrument Programming : Simulated Configuration Files

Simulated Configuration Files

You can use the UltraWave 12G instruments offline by forcing IG-XL to map the UltraWave subsystem. Do this by creating a SimulatedConfig.txt file and placing the file in any of the following locations:

- Folder containing the IG-XL workbook
- `\tester` folder: C:\Program Files (x86)Teradyne\IG-XL\<version>\tester
- `\bin` folder: C:\Program Files (x86)\Teradyne\IG-XL\<version>\bin

The following SimulatedConfig.txt file maps the UltraWave 12G boards in slots 2, 3 and 4. This configuration has an MWSynthRef board in addition to the MWSynthLO board and only one MWMeasure board.

```
<Version>
1
</Version>
```

<TEST_HEAD TH 1>

#slot[subslot]	Type	idprom (type serial rev company)
2.0	MWMeasure	805-041-00 0000000 0000-A 0000
	MWMeasure	939-399-00 0000000 0000-A 0000
	FrontEnd1	599-021-00 0000000 0000-A 0000
	FrontEnd2	599-021-00 0000000 0000-A 0000

	FrontEnd3	599-021-00 0000000 0000-A 0000
	FrontEnd4	599-021-00 0000000 0000-A 0000
	SplitterAtten1	599-022-00 0000000 0000-A 0000
	SplitterAtten2	599-022-00 0000000 0000-A 0000
	Receiver1	939-401-00 0000000 0000-A 0000
	Receiver2	939-401-00 0000000 0000-A 0000
	Digitizer	939-400-00 0000000 0000-A 0000
	PowerDistribution	939-411-00 0000000 0000-A 0000
	PowerBoard	939-402-00 0000000 0000-A 0000
3.0	MWSynthLO	805-043-00 0000000 0000-A 0000
	MWSynthLO	939-399-00 0000000 0000-A 0000
	PLLFeedback1	939-405-00 0000000 0000-A 0000
	PLLFeedback2	939-405-00 0000000 0000-A 0000
	OutputDivider1	599-024-00 0000000 0000-A 0000
	OutputDivider2	599-024-00 0000000 0000-A 0000
	AWG	939-390-00 0000000 0000-A 0000
	Modulator	599-025-00 0000000 0000-A 0000
	BaseBand	939-403-00 0000000 0000-A 0000
	PowerBoard	939-402-00 0000000 0000-A 0000
4.0	MWSynthRef	805-044-00 0000000 0000-A 0000
	MWSynthRef	939-399-00 0000000 0000-A 0000
	PLLFeedback1	939-405-00 0000000 0000-A 0000
	PLLFeedback2	939-405-00 0000000 0000-A 0000
	AWG	939-390-00 0000000 0000-A 0000
	Modulator	599-025-00 0000000 0000-A 0000
	BaseBand	939-403-00 0000000 0000-A 0000
	RefSourceFanout	599-026-00 0000000 0000-A 0000
	OutputDivider	599-024-00 0000000 0000-A 0000
	PowerBoard	939-402-00 0000000 0000-A 0000
24.0	SupportBoard	974-220-12 0000000 0000-A 0000
	SupportBoard	956-354-00 0000000 0000-A 0000
62.0	DistributionBoard	956-355-00 0000000 0000-A 0000
	DistributionBoard	956-355-00 0000000 0000-A 0000

```
</TEST_HEAD TH 1>
```

The simulated configuration file also requires a MWSynth cable section:

```
<CABLE MWSYNTH 1>
#Sythesizer Channel      Measure Channel
#<THSlot>.<MWSynth Out>  <THSlot>.<MWMeasure In>
3.LO                     2.LO
3.CW/ModSrc              2.CW/ModSrc1
4.CW/ModSrc              2.CW/ModSrc2
</CABLE MWSYNTH 1>
```

To use the [UltraWave 12G instruments offline](#), the SimulatedConfig.txt file must include every board needed in the IG-XL workbook. The SimulatedConfig.txt file represents the state of the offline tester. To use the UltraWave 12G instruments online, you need not edit any files. IG-XL automatically maps all boards in the tester.
For more information, see [Tester Configuration](#).

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UltraWave 12G Instrument Programming : IG-XL Programming for UltraWave 12G Instruments

IG-XL Programming for UltraWave 12G Instruments

You create MW tests (test functions using MW logical instruments) using the VB editor to code VBT functions in VB modules. MW tests become part of a test program when you include them in a worksheet in an IG-XL workbook. MW tests require the following data, typically entered using DataTool:

- | | |
|--|--|
| | <ul style="list-style-type: none">DUT pin data in a Pin Map sheet. See Programming the Pin Map. |
| | <ul style="list-style-type: none">Tester channel data in a Channel Map sheet. See Programming the Channel Map. |
| | <ul style="list-style-type: none">Test instance data in a Test Instances sheet |
| | <ul style="list-style-type: none">Test names and execution flow control data in the Flow Table sheet |

For more information about IG-XL workbooks, worksheets, and data, see About DataTool.

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UltraWave 12G Instrument Programming : IG-XL Programming for UltraWave 12G Instruments : Programming the Pin Map

Programming the Pin Map
A pin map describes the device pins by assigning a pin type to each device pin name. For each RF DUT pin in a new or existing Pin Map sheet:

1. Type a name for the device pin, such as RF_in, in the Pin Name column.
2. Select Analog as the pin type, in the Type column.

For more information, see Pin Map Sheet in the DataTool section.

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UltraWave 12G Instrument Programming : IG-XL Programming for UltraWave 12G Instruments : Programming the Channel Map

Programming the Channel Map

A Channel Map sheet assigns individual device pin names to test system channels and power supplies. For DIB connections (MW ports, Pogo pins) and signal names for the MW subsystem and instruments, see [User View of UltraWave 12G SMPM Connector Blocks](#), [User View of UltraWave 12G DIB Pad Patterns](#), and [UltraWave 12G Signal Descriptions](#).

The UltraWave 12G has two channel types:

- | | |
|--|---|
| | <ul style="list-style-type: none">MW: Identifies a bidirectional channel (port) on a MW Measure board. |
| | <ul style="list-style-type: none">MWSource: Identifies a DUT LO port or DUT Clock reference pins on a MW Synthesizer Reference board. |

The MWSource channel type supports only the MWSource logical instrument. IG-XL allows you to use the MWSource channel type on a bidirectional port, but then you cannot use any other MW logical instruments. There is no advantage to using the MWSource channel type on a bidirectional port. Use the MWSource channel type with the MWSource logical instrument to program the DUT LO ports and the DUT Clock pins.

To program the MW channel type in a channel map, choose the corresponding instrument type:

- | | |
|--|--|
| | <ul style="list-style-type: none">All MW Instruments Except DUT LO and DUT Clock |
| | <ul style="list-style-type: none">DUT LO and DUT Clock |

All MW Instruments Except DUT LO and DUT Clock

For each RF pin in a new or existing channel map, for all MW instruments except a DUT LO or DUT Clock:

1. [Select](#) MW as the tester channel type, in the Type column.
2. [Enter a Pogo pin name or signal name to the](#) Tester Channel Site column to assign a MW tester channel to the pin.

DUT LO and DUT Clock

For each RF pin in a new or existing channel map for a DUT LO or DUT Clock:

1. [Select](#) MWSource as the tester channel type, in the Type column.
2. [Enter a Pogo pin name or signal name to the](#) Tester Channel Site column to assign a MW tester channel to the pin.

The figures below show a channel map for source, measure, and DUT clock pins using the Signal and Pogo names.

Channel Map



DIB ID:

View Mode: [Signal](#)

Site Replication:

Device Under Test		Tester Channel				
Pin Name	Package Pin	Type	Site 0	Site 1	Site 2	Site 3
rf_source		MW	4.MW4A	4.MW1A	2.MW4A	2.MW1A
rf_measure		MW	4.MW3C	4.MW2C	2.MW3C	2.MW2C
ref_clk		MWSource	10.RefClk1D	10.RefClk1C	10.RefClk1B	10.RefClk1A

UltraWave 12G Channel Map Using Signal Names

Channel Map



DIB ID:

View Mode: [Pogo](#)

Site Replication:

Device Under Test		Tester Channel				
Pin Name	Package Pin	Type	Site 0	Site 1	Site 2	Site 3
rf_source		MW	4.smpm15	4.smpm1	2.smpm15	2.smpm1
rf_measure		MW	4.smpm13	4.smpm7	2.smpm13	2.smpm7
ref_clk		MWSource	10.b2	10.c1	10.d2	10.e1

UltraWave 12G Channel Map Using Pogo Names

For more information, see [Channel Map Sheet](#) in the DataTool section.

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UltraWave 12G Instrument Programming : Using UltraWave 12G in Concurrent Testing

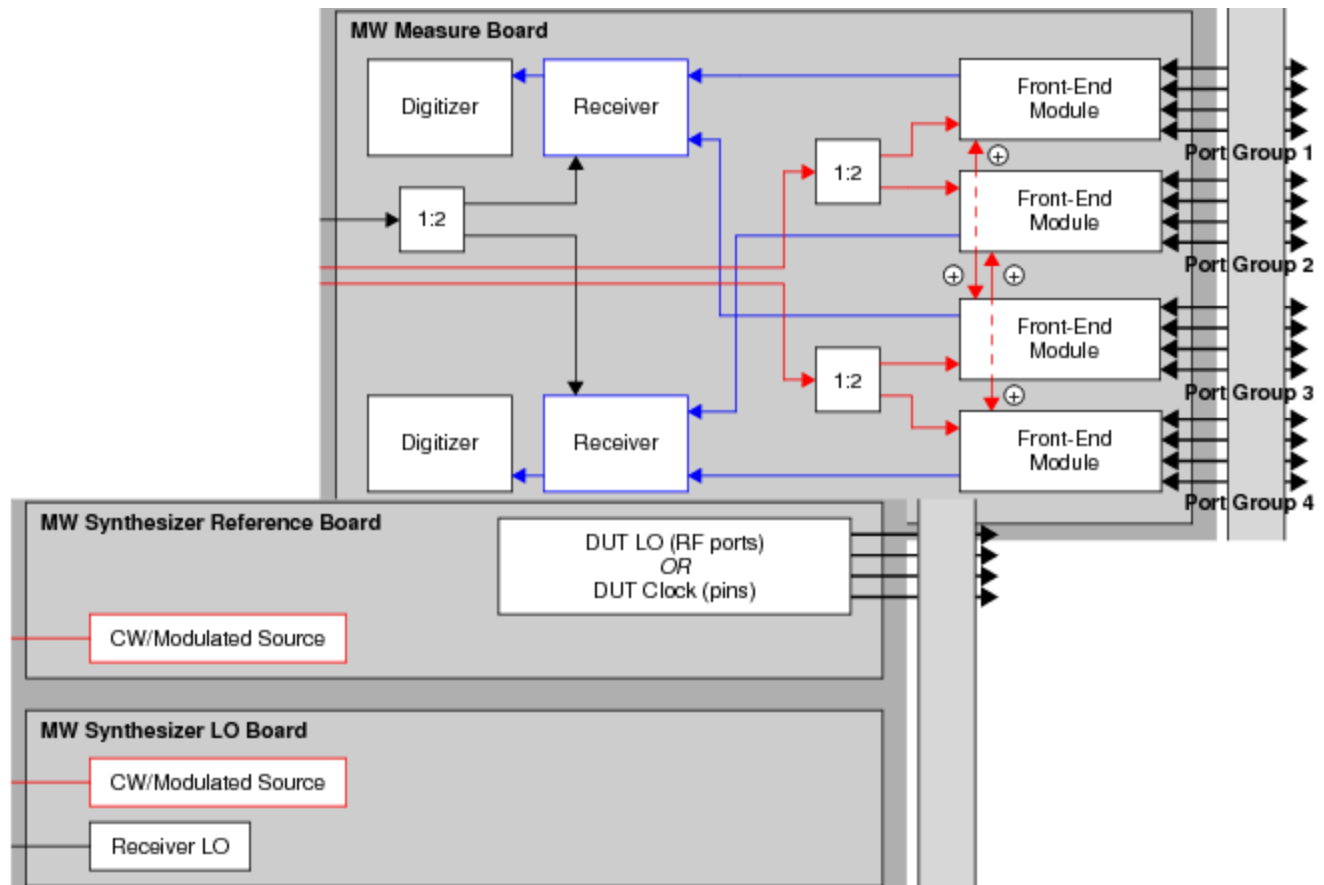
Using UltraWave 12G in Concurrent Testing

To support concurrent testing, you can assign each UltraWave hardware block to unique time domains. Concurrent features for UltraWave include:

- A MW Measure board can belong to up to four distinct time domains. This provides support for up to 16 time domains on four MW Measure boards.
- A MW Synthesizer LO or MW Synthesizer Reference board can belong to up to two distinct time domains
- Each UltraWave hardware block can be controlled by unique time domains simultaneously using Sync-Link. Features include:
 - Independent access to Sync-Link
 - Independent sample clock programming
 - Independent clock synchronization with other UltraWave hardware blocks or with digital clocks
 - Independent control by digital pattern
 - Independent capture memory and triggers

- Multiple hardware blocks can be grouped into a common time domain
- UltraWave 12G fully supports flow domains and asynchronous pattern starts

The following diagram shows the granularity of time domain programming for UltraWave boards and logical instruments. Each white box in the diagram is a hardware block. For example, each Front End Module on a Measure board can be assigned to distinct time domains (up to four per board).



UltraWave Time Domain Granularity

For more information, see [About Concurrent Test](#).

Duplex MTD Example

The following example shows the elements that are included in an UltraWave test program for multiple time domains. This example describes a program that uses an MWSource in one time domain for the RX side of a device and an MWReceiver in another time domain for the TX side of the device.

The following figures show an example pin map and channel map.

Pin Map

Group Name	Pin Name	Type
	Source	Analog
	Measure	Analog
	Dig1	I/O
	Dig2	I/O
MWSourceOnly	Dig1	TimeDomain
MWSourceOnly	Source	
MWReceiveOnly	Dig2	TimeDomain
MWReceiveOnly	Measure	

Example Pin Map

Channel Map

DIB ID:

View Mode:

Device Under Test		Tester Channel		
Pin Name	Package Pin	Type	Site 0	Site 1
Source		MW	2.MW1A	2.MW2A
Measure		MW	2.MW3A	2.MW4A
Dig1		I/O	1.ch16	1.ch17
Dig2		I/O	1.ch128	1.ch129

Example Channel Map

For most UltraWave 12G instruments, mapping of the front end module used for each channel is not complex since the module is the channel map number. So, in this example:

- Pin Source uses front end module 1 (MW1A) for Site 0 and module 2 (MW2A) for Site 1.
- Pin Measure uses front end module 3 (MW3A) for Site 0 and module 4 (MW4A) for Site 1.

However, mapping of MWSource and MWReceiver is not as obvious. This figure below shows the best way to partition resources, mapping odd sites to the even front end modules and even sites to the odd modules. This organization expands in the same way for more sites and measure boards.

MW Channel	Site 0				Site 1			
	1		3		2		4	
	Src X	Rcv X	Src Y	Rcv X	Src X	Rcv Y	Src Y	Rcv Y
A								Concurrency Group 1
B								
C								Concurrency Group 2
D								

Mapping of MWSource and MWReceiver for Concurrent Test

You must also include VBT code to OnProgramLoaded:

```
TheHdw.MW.EnableIntraSlotMTD = True
```

So, the function would look something like this:

```
Function OnProgramLoaded()
```

```
    On Error GoTo errHandler
```

```
    ' Put code here
```

```
    TheHdw.MW.EnableIntraSlotMTD = True
```

```
Exit Function
```

errHandler:

```
    HandleExecIPError "OnProgramLoaded"
```

```
End Function
```

For more information, see [EnableIntraSlotMTD](#) in the VBT Syntax Reference.

If you want to control pattern interlocking while a pattern is running, add the following to OnProgramStarted:

```
TheHdw.MW.POPInterlock = tlMWPOPIterlockEnable
```

So, the function would look something like this:

```
Function OnProgramStarted()
```

```
    On Error GoTo errHandler
```

```
    ' Put code here
```

```
    TheHdw.MW.POPInterlock = tlMWPOPIterlockEnable
```

```
Exit Function
```

errHandler:

```
    HandleExecIPError "OnProgramStarted"
```

```
End Function
```

For more information, see [POPInterlock](#) in the VBT Syntax Reference.

The program also includes patterns defined as follows:

- One time domain would contain tester source instruments (such as ModSrc) for the RX side of the DUT
- One time domain would contain tester receive instruments (such as MWReceiver) for the TX side of the DUT

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[UltraWave 12G Instrument Programming](#) : [UltraWave 12G Pattern Control](#)

[UltraWave 12G Pattern Control](#)

You can control an UltraWave 12G logical instruments using pattern microcodes, allowing changes to instrument states and parameters during a program. You can obtain repeatable timing by programming an instrument using pattern microcodes and Signals programming.

You create a pattern independently of the test program using the PatternTool or a text editor.

Patterns are composed of a series of numbered digital vectors. These digital vectors define the data for each channel listed in the pattern, as well as the pattern microcode used for controlling the instrument during pattern execution.

This section includes general topics on UltraWave 12G pattern control. The following topics are available:

- [Using UltraWave 12G Instruments with POP](#)

- [Understanding UltraWave Pattern Interlocking](#)

For more information on pattern control for UltraWave 12G instruments, see:

- [MWModulatedSource Pattern Control](#)

- [MWMultiToneSource Pattern Control](#)

- [MWNoiseSource Pattern Control](#)

- [MWReceiver Pattern Control](#)

- [MWSource Pattern Control](#)
- [UltraWave 12G VNA Pattern Control](#)

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UltraWave 12G Instrument Programming : UltraWave 12G Pattern Control : Using UltraWave 12G Instruments with POP

Using UltraWave 12G Instruments with POP

Pattern oriented programming (POP) is a test programming style that has many benefits including measurement stability, test reliability, and test time. This type of programming can be used with UltraWave logical instruments.

POP uses pattern memory and Digital instrument boards that are present on all UltraFLEX testers. The PatGens on the Digital instruments control the application of patterns and can respond to DUT and instrument conditions in real time. The PatGens also maintain synchronization with each other through the SyncLink. Pattern memory contains not only digital 1s and 0s but also analog instrument opcodes. The analog instrument opcodes perform functions like applying a PSet, performing a strobe, or starting a signal using Start.

Because of this test system architecture, once a pattern is started, the instruments can handle all aspects of a test: applying conditions, performing measurements, and reacting to changing test conditions. And because timing and synchronization are handled by the hardware, test repeatability and precision is superior to what the test computer can achieve.

While a pattern is the executing a test, the test computer can communicate with tester resources through the test system bus or can process data.

When using an UltraWave logical instrument (MWModulatedSource or other instrument), any vector that contains a POP microcode (either LoadSettingsConnect or Disconnect) must be fully defined across all defined logical instrument instances. IG-XL returns an error if you do not follow this guideline.

Reapplying the same Signal to a given logical instrument instance results in the hardware NOPing the application. (This prevents glitches.)

The following figure shows an example pattern file (UltraWavePOP.PAT) in PatternTool that uses POP microcodes and several UltraWave logical Instruments. Note the following:

- For vectors with POP microcodes, the example pattern demonstrates the idea of defining the state for all logical instruments. Vectors 0, 7, 14, and 20 contain POP microcodes.
- Vector 14 includes a comment that this vector will not cause a glitch on the MWSource pin (as the same Signal is reapplied). This same concept applies to all sourcing logical instruments any time the same Signal is reapplied.

UltrawavePOP.PAT (vm_UltraWavePOP - VM)									
TSet	Command	Label	Vector	Source (MWModulatedSource)	Source (MWMultitoneSource)	Measure (MWReceiver)	Source (MWSource)	Dig1	Comment
Basic	start_label UltrawavePOP		0	LoadSettingsConnect GSM	Disconnect	LoadSettingsConnect GSM	Disconnect	0	
Basic	repeat 10000		1					0	1mS wait time
Basic			2	Start GSM				0	
Basic	repeat 100		3					0	10uS wait time
Basic			4			Trig GSM		0	
Basic	repeat 10000		5					0	1mS wait time
Basic			6					0	
Basic			7	Disconnect	Disconnect	LoadSettingsConnect WCDMA	LoadSettingsConnect WCDMA	0	
Basic	repeat 10000		8					0	1mS wait time
Basic			9					0	
Basic	repeat 100		10					0	10uS wait time
Basic			11			Trig WCDMA		0	
Basic	repeat 10000		12					0	1mS wait time
Basic			13					0	
Basic			14	Disconnect	Disconnect	LoadSettingsConnect WCDMA2	LoadSettingsConnect WCDMA2	0	No glitch created on MWSource, same
Basic	repeat 10000		15					0	1mS wait time
Basic			16					0	
Basic	repeat 100		17					0	10uS wait time
Basic			18			Trig WCDMA2		0	
Basic	repeat 10000		19					0	1mS wait time
Basic			20	Disconnect	Disconnect	Disconnect	Disconnect	0	
Basic	repeat 2000		21					0	200uS wait time
Basic			22					0	
Basic			23					0	

Vertical: 65 Vectors

Vectors 0 to 23

POP Example Pattern Using UltraWave Instruments

The following is the ASCII code for the example pattern.

```
instruments = {  
    Source:MWModulatedSource 1;  
    Source:MWMultitoneSource 1;  
    Measure:MWReceiver 1;  
    Source:MWSource 1;  
}  
  
vm_vector  
vm_UltraWavePOP          ( $tset, Dig1)  
{  
    start_label Ultrawave_POP:  
    (Source:MWModulatedSource = LoadSettingsConnect GSM)  
    (Source:MWMultitoneSource = Disconnect)
```

```

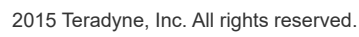
(Measure:MWRceiver = LoadSettingsConnect GSM)
(Source:MWSource = Disconnect)
    > Basic 0 ;
repeat 10000    > Basic 0 ; //1ms wait time
(Source:MWModulatedSource = Start GSM)
    > Basic 0 ;
repeat 100    > Basic 0 ; //10us wait time
(Measure:MWRceiver = Trig GSM)
    > Basic 0 ;
repeat 10000    > Basic 0 ; //1ms wait time
    > Basic 0 ;
(Source:MWModulatedSource = Disconnect)
(Source:MWMultitoneSource = Disconnect)
(Measure:MWRceiver = LoadSettingsConnect WCDMA)
(Source:MWSource = LoadSettingsConnect WCDMA)
    > Basic 0 ;
repeat 10000    > Basic 0 ; //1ms wait time
    > Basic 0 ;
repeat 100    > Basic 0 ; //10us wait time
(Measure:MWRceiver = Trig WCDMA)
    > Basic 0 ;
repeat 10000    > Basic 0 ; //1ms wait time
    > Basic 0 ;
(Source:MWModulatedSource = Disconnect)
(Source:MWMultitoneSource = Disconnect)
(Measure:MWRceiver = LoadSettingsConnect WCDMA2)
(Source:MWSource = LoadSettingsConnect WCDMA)
    > Basic 0 ; //No glitch created on
                //MWSource because same
                //Signal (WCDMA) is
                //reapplied.
repeat 10000    > Basic 0 ; //1ms wait time
    > Basic 0 ;
repeat 100    > Basic 0 ; //10us wait time
(Measure:MWRceiver = Trig WCDMA2)
    > Basic 0 ;
repeat 10000    > Basic 0 ; //1ms wait time
(Source:MWModulatedSource = Disconnect)

```

(Source:MWSource = Disconnect)

```
> Basic 0 ;
```

```
halt
}
```



UW12GSubsys_Prog.4.9



UltraWave 12G Instrument Programming : UltraWave 12G Pattern Control : Understanding UltraWave Pattern Interlocking

Understanding UltraWave Pattern Interlocking

By default, hardware for UltraWave 12G instruments is set to listen to programming commands that are initiated from VBT and that originate from the host computer. When a digital pattern is burst that contains UltraWave microcodes, such as LoadSettingsConnect, the hardware is activated to also listen for instructions that originate from the digital pattern generator which are sent through the Instrument Sync Link (ISL) bus.

While an UltraWave instrument is set to listen for ISL instructions, IG-XL locks out all commands originating from VBT, and a run-time error is generated if you attempt to program the UltraWave instrument from VBT while the pattern is running. This pattern interlocking is done to avoid collisions in the hardware in the case where instructions from different sources occur too close together for the hardware to react properly to both instructions.

The lockout starts when a digital pattern containing LoadSettingsConnect or Disconnect microcodes is started. The lockout is released at end of a Digital.Patgen.Haltwait VBT statement indicating that the pattern burst is complete or when an explicit Patgen.Halt method is executed. At times, however, a test program may be written such that VBT commands are to be issued to an UltraWave instrument while a pattern is running. For example, you might need to issue commands while a pattern is looping on a wait for CPU flag. In such cases, you can override the pattern lockout to allow VBT programming to occur. The syntax is:

```
TheHdw.MW.PatternMicroCodeControl = tIMWPatternMicroCodeControlOff
```

This lockout is controlled per time domain for UltraWave instruments.

Once the needed VBT commands are issued, restore .PatternMicroCodeControl to tIMWPatternMicroCodeControlOn before continuing pattern execution.

Note that changing the .PatternMicroCodeControl property only toggles the VBT interlock mechanism. It does not disable ISL commands. Collisions are still possible when

PatternMicroCodeControl is set to tlmwPatternMicroCodeControlOff and the digital pattern is still running and executing microcodes. Take care when toggling .PatternMicroCodeControl such that this does not occur.

For more information, see [PatternMicroCodeControl](#) in the VBT Syntax Reference.

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UltraWave 12G Theory of Operation

UltraWave 12G Theory of Operation

The UltraWave 12G subsystem provides microwave instrument hardware functions on boards and splitter modules in the test head. The hardware functions are allocated, configured, and connected in different ways to provide MW instrument capabilities. These software configured MW instruments are called logical instruments. The operation of individual logical instruments depends on the operation of the subsystem functions.

The following topics are available:

- [UltraWave 12G Subsystem Architecture](#)
- [UltraWave 12G Hardware Functions](#)
- [UltraWave 12G Logical Instruments and Hardware Functions](#)
- [User View of UltraWave 12G SMPM Connector Blocks](#)
- [User View of UltraWave 12G DIB Pad Patterns](#)
- [UltraWave 12G Signal Descriptions](#)

For information on UltraWave 12G instrument performance specifications and standards, see [UltraWave 12G Specifications](#).

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UltraWave 12G Theory of Operation : UltraWave 12G Subsystem Architecture

UltraWave 12G Subsystem Architecture

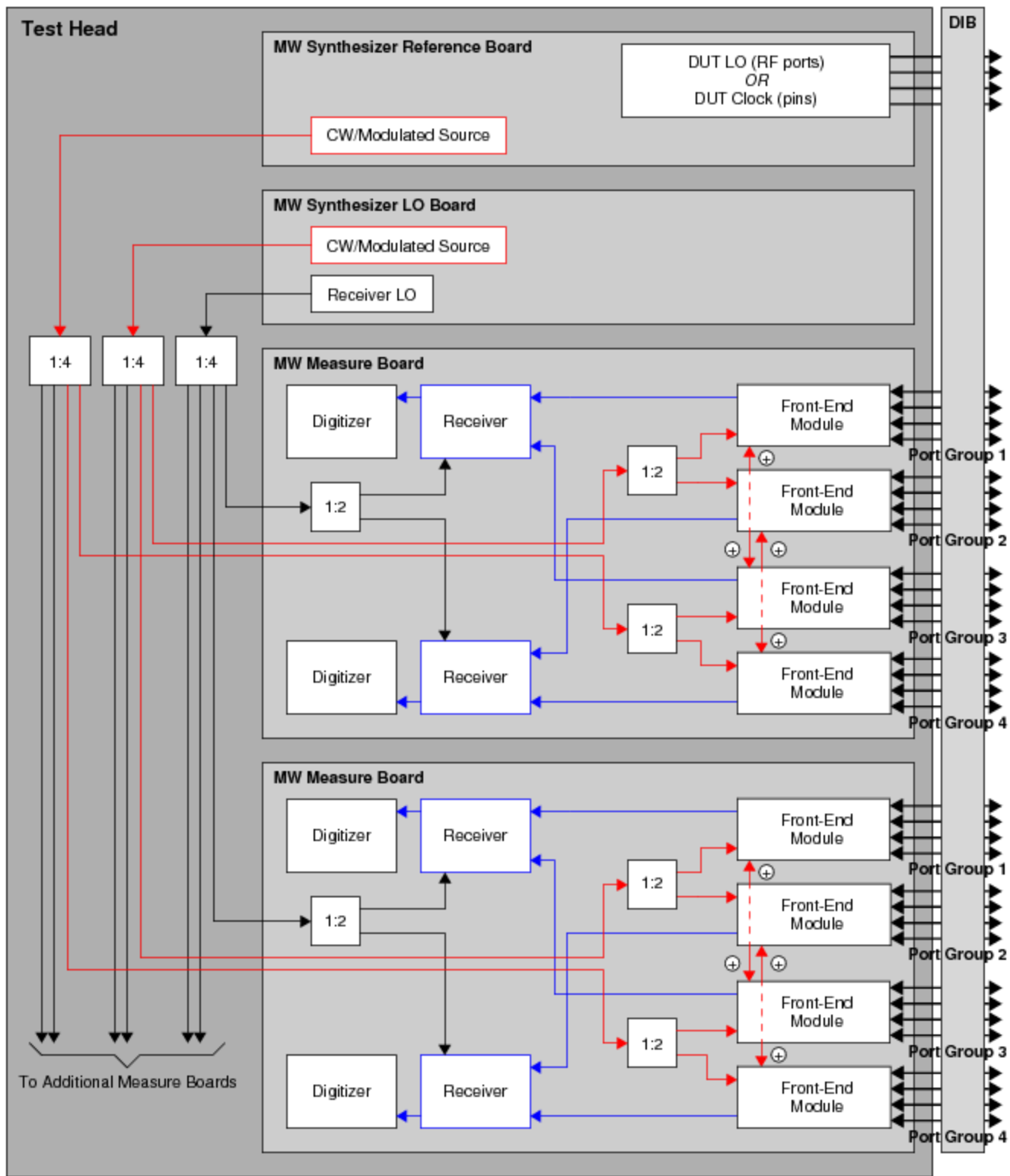
All hardware functions in the UltraWave 12G subsystem are on boards and splitter modules in the test head. The figure shows the basic, high-level hardware functions and signal connections for a typical configuration having 32 RF ports: two MW Measure boards and two MW Synthesizer boards (one MW Synthesizer LO board and one MW Synthesizer Reference board).

As shown, the MW Synthesizer LO board has an LO synthesizer for the down-converting receivers on one or more MW Measure boards, plus a synthesizer and modulator for use as a CW or modulated signal generator for the DUT. The MW Synthesizer Reference board also has a synthesizer and modulator for use as a signal generator, but the second synthesizer provides either a DUT LO (through RF ports) or a DUT clock (through spring-loaded pins).

The UltraWave 12G subsystem is expandable for high pin-count devices and multiple sites. Each MW Synthesizer LO or MW Synthesizer LO/MW Synthesizer Reference pair is connected to 1:4 splitter modules in the test head which can in turn support up to four measure boards (64 RF ports). A second MW Synthesizer LO/MW Synthesizer Reference pair can be added, for up to 96 ports maximum. For more information, see Supported UltraWave 12G Configurations.

For more information on UltraWave 12G architecture, see the following topics:

- Detailed Architecture and Resource Mapping
- Shared Hardware Functions
- Supported UltraWave 12G Configurations



UltraWave 12G Subsystem Hardware Architecture

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UltraWave 12G Theory of Operation : UltraWave 12G Subsystem Architecture : Detailed Architecture and Resource Mapping

Detailed Architecture and Resource Mapping

The UltraWave 12G has a fanout architecture that allows increasing port count by adding only MW Measure boards. The fanout ratio is 1:16 when a single CW/Modulated source is connected to both source splitter inputs on a Measure board and 1:8 when two CW/Modulated sources are connected to a Measure board. When a single source is used, the source is available at all ports. However, when two sources are used, each source is available at half of the ports.

UltraWave 12G has a single LO source, so that the LO is common to all receivers.

Each Measure board has two receivers, each with a digitizer. Each digitizer can connect to half of the ports.

As shown in UltraWave 12G Subsystem Hardware Architecture, a Measure board includes:

- A single LO source input for both receivers, going into a 1:2 splitter
- Two receivers, each connected to a dedicated digitizer
- Two signal source inputs, each going into a coarse attenuator/1:2 splitter
- Four front-end modules (MW universal instrument channels), each including:
 - Source signal input from a coarse attenuator/1:2 splitter
 - Receive signal output

- Multitone source add path signal input/output

- 1:4 bidirectional mux to RF DIB ports

☑ **Note:**

The bidirectional “add paths” for the multitone sources are designated by the $\oplus$ symbol. The front-end modules are connected only at signal ends with arrows; the dotted line signal paths are not connected to the front-end modules they pass through (“behind”) in the figure.

A fanout architecture has rules for mapping resources (hardware functions) to ports for the supported configurations. These rules affect DIB port connections and sharing of hardware functions. The rules depend on the hardware configuration:

- Single Source Configuration Resource Mapping Rules

- Dual Source Configuration Resource Mapping Rules

For more information, see Supported UltraWave 12G Configurations.

Single Source Configuration Resource Mapping Rules

In single source configurations, the two outputs from the 1:4 splitter can be connected to the two measure board source inputs.

The following are resource mapping rules for a single source configuration:

- The receiver LO input for both the first and second receiver can connect only to the splitter for the single LO frequency input to the measure board.

- The first and third front-end modules can connect only to the first receiver and not to the second receiver.

- The second and fourth front-end modules can connect only to the second receiver and not to the first receiver.

- The first and third front-end modules can connect only to each other through the add path as parts of a multitone source instrument and not to the second or fourth front-end modules.

- The second and fourth front-end modules can connect only to each other through the add path as parts of a multitone source instrument and not to the first or third front-end modules.

Dual Source Configuration Resource Mapping Rules

A quad source configuration is two copies of a dual source configuration, so the dual source configuration resource mapping rules apply.

The following are resource mapping rules for a dual source configuration:

- The receiver LO input for both the first and second receiver can connect only to the splitter for the single LO frequency input to the Measure board.
- The first source input can connect only to the first and second front-end modules through a splitter/attenuator, and not to the third and fourth front-end modules.
- The second source input can connect only to the third and fourth front-end modules through a splitter/attenuator, and not to the first and second front-end modules.
- The first and third front-end modules can connect only to the first receiver and not to the second receiver.
- The second and fourth front-end modules can connect only to the second receiver and not to the first receiver.
- The first and third front-end modules can connect only to each other through the add path as parts of a multitone source instrument and not to the second or fourth front-end modules.
- The second and fourth front-end modules can connect only to each other through the add path as parts of a multitone source instrument and not to the first or third front-end modules.



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UltraWave 12G Theory of Operation : UltraWave 12G Subsystem Architecture : Shared Hardware Functions

Shared Hardware Functions

The UltraWave 12G subsystem architecture requires MW logical instruments to share certain hardware functions:

1.

Each measure board (up to four) shares the synthesizers for the receiver LO and CW/modulator generator functions with the other measure boards through 1:4 splitters. Because the synthesizers are shared:

–

All active receiver ports must use the same frequency settings.

–

All active source ports using an individual CW/modulated source must use the same frequency settings and CW/modulated signal content. Fine attenuation level settings can be different.

2.

Each front-end module on a measure board shares a synthesizer for a single CW/modulated source (on either an LO source board or a DUT ref source board) with another front-end module through a 1:2 splitter with coarse leveling (see pairs on the diagram).

Because the synthesizers are shared, all active source ports on the pair of front-end modules must use the same frequency settings and CW/modulated signal content (see point 1, above) and power level.

3.

Each front-end module can route a source that is providing a component of a multitone signal to one other front-end module (see pairs on diagram marked with ⊕). The source and receive

paths share the step attenuator on a front-end module.

Because the step attenuator is shared, a front-end module routing a multitone waveform component to the corresponding paired front-end module (where the component is added and connected to a port) can only route a signal from a port on it to the receiver using the coupled RF input path.

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UltraWave 12G Theory of Operation : UltraWave 12G Subsystem Architecture : Supported
UltraWave 12G Configurations

Supported UltraWave 12G Configurations

The capabilities of the supported UltraWave 12G configurations depend on:

- The number of Synthesizer boards: Determines the number of independent sources.
- The number of Measure boards: Determines the number of ports, up to the limit imposed by the number of Synthesizer boards or the DIB cabling limit.

Supported Configurations

Supported UltraWave 12G Configurations

Synthesizer Boards	Independent Sources	RF Ports
1 MW Synthesizer LO	1 CW/Modulated Source	16, 32
1 MW Synthesizer LO + 1 MW Synthesizer Reference	2 CW/Modulated Sources	16, 32, 48, 64
2 MW Synthesizer LO + 2 MW Synthesizer References	4 CW/Modulated Sources	32, 64, 96 Maximum ports limited by DIB cabling space.

A minimum subsystem configuration (16 RF ports) has two boards:

- One MW Synthesizer LO board

- [One MW Measure board](#)

In this configuration, only one CW/modulated source is available. The single source can be connected to the second 1:2 splitter on the measure board using an output of the 1:4 splitter module. This allows the source to be available at all ports.

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UltraWave 12G Theory of Operation : UltraWave 12G Hardware Functions

UltraWave 12G Hardware Functions

The UltraWave 12G subsystem can be represented by hardware function blocks that support the MW logical instruments. The logical instruments connect the functions in a number of ways for specific source and measure capabilities. The functions and connections are described in the following topics:

- | | |
|---|--|
| • | UltraWave 12G Subsystem Function Block Diagram |
| • | UltraWave 12G Function Descriptions |

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UltraWave 12G Theory of Operation : UltraWave 12G Hardware Functions : UltraWave 12G Subsystem Function Block Diagram

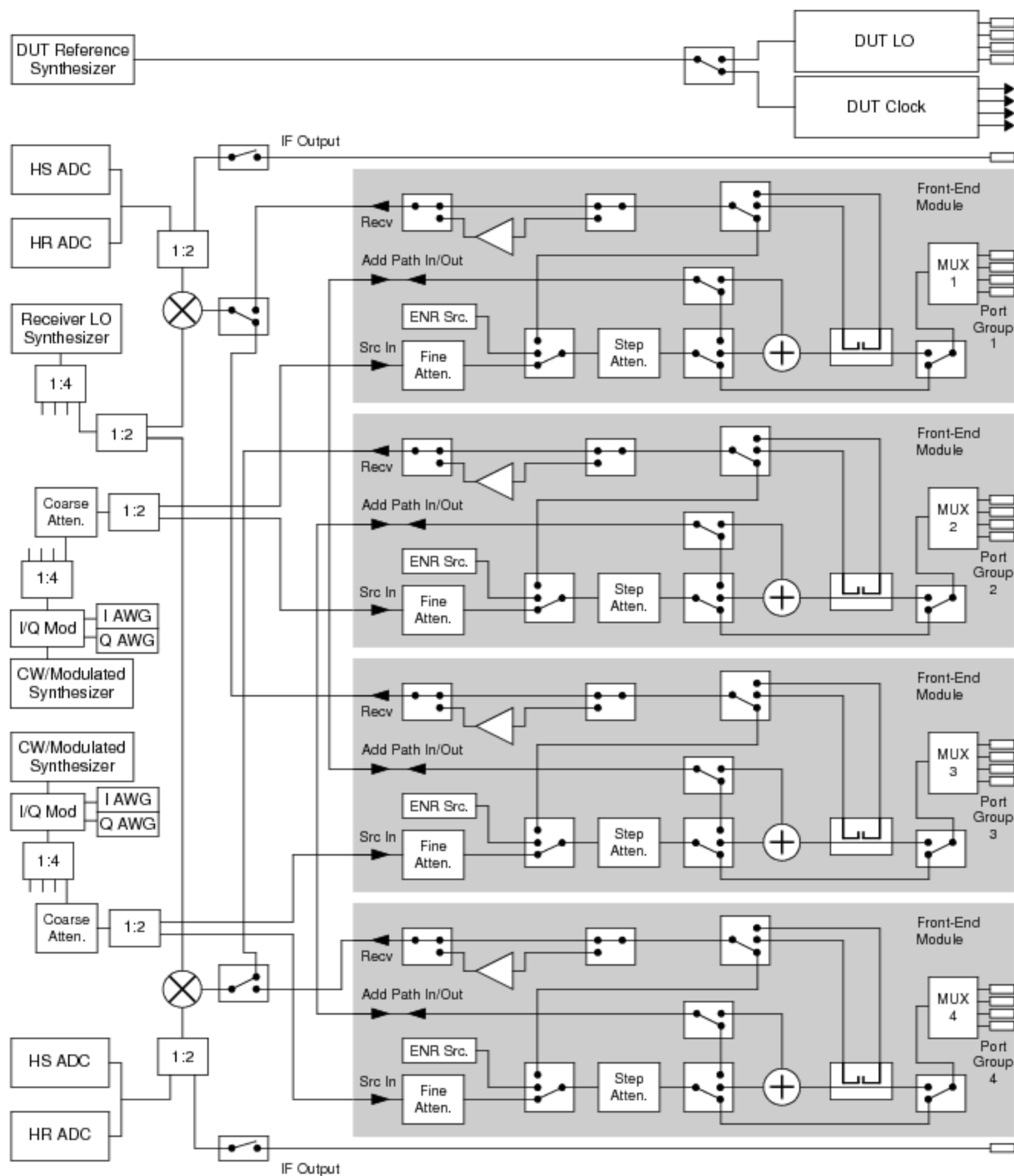
UltraWave 12G Subsystem Function Block Diagram

The figure shows UltraWave 12G function blocks and switch functions that connect MW functions in various ways to create and control logical instruments. The switch arrangements represent how the functions can be logically connected and how signals flow, but are not necessarily identical to physical hardware switches. Any MW logical instrument, and any valid combination of logical instruments operating simultaneously, can be represented as a group of functions and switch settings on the diagram.

The block diagram shows an example system with Synthesizer LO board, a Synthesizer Reference board, and one Measure board. All DIB connections are RF ports, except the connections for the DUT clock.

For more information on the function blocks, see UltraWave 12G Function Descriptions. For DIB connections (MW ports, Pogo pins) and signal names for the MW subsystem and instruments, see:

- User View of UltraWave 12G SMPM Connector Blocks
- User View of UltraWave 12G DIB Pad Patterns
- UltraWave 12G Signal Descriptions



UltraWave 12G Function Block Diagram: 16 Universal Ports plus 4 DUT Clock Ports or Pins

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[UltraWave 12G Theory of Operation](#) : [UltraWave 12G Hardware Functions](#) : [UltraWave 12G Function Descriptions](#)

UltraWave 12G Function Descriptions

The UltraWave 12G Subsystem includes the following functions:

1:2	1:2 power splitter. Splits the following signals: • Receiver LO for two receivers on a measure board • Coarse-attenuated source from CW/Modulated synthesizer for two source channels on a measure board • Receiver I/F output between the digitizer and DIB I/F output
1:4	1:4 power splitter. Splits the following signals: • Receiver LO output for one to four measure boards • CW/Modulated synthesizer output for sources on one to four measure boards
Coarse Attenuator	Attenuator with large (20dB) steps. Provides approximate level to each source channel. Exact level is set by the fine and step attenuators.

CW/Modulated Synthesizer	Synthesizer for CW (continuous wave or sine) signals and modulated signals.
DUT Reference Synthesizer	Synthesizer for a DUT LO or a DUT Clock.
DUT LO	LO clock for a DUT. Available to the DIB at four RF (subminiature coax) ports.
DUT Clock	Reference clock for a DUT or DIB circuit. Available to the DIB at four Pogo (spring-loaded) pins.
ENR Source	Calibrated wideband noise source.
Fine Attenuator	Attenuator with very fine steps. Provides fine adjustments to adjust source signals to specified level accuracy. Can also adjust levels at ports to different levels within a coarse range.
HR ADC	High resolution analog-to-digital converter (500kHz-800kHz BW, 24 bit).
HS ADC	High speed analog-to-digital converter (500kHz-45MHz BW, 16 bit).
Pogo pin	High frequency tester pin. Spring-load tester pin on sidewalk 4 at the DIB interface. Contrast to RF port which is a subminiature coax connector.
I AWG	Arbitrary waveform generator for the I (in-phase) modulation waveform data input to the modulated source.
IF output	IF output of the receiver down-converter. Switched and available to the DIB at an RF (subminiature coax) port.
I/Q Mod	I/Q modulator for the modulated source.
LF ADC	Low frequency analog-to-digital converter.
LNA 	Low-noise amplifier in receive signal path. Bypassed when not needed.
MUX N	Bidirectional multiplexer for each MW instrument channel. All channels, multiplexers, and RF ports on a measure board have the same performance.
Q AWG	Arbitrary waveform generator for the Q (quadrature) modulation waveform data input to the modulated source.
Receiver 	RF receiver (mixer/downconverter).
Receiver LO Synthesizer	Synthesizer of LO signal for receivers.

Reflected Coupler	Signal coupler for reflected waves. Required for s-parameter measurements.
RF Port	Bidirectional RF connection at the DIB interfaced. Multiplexed 1:4 for source and receive channels.
Signal Adder 	Adds (combines) two RF signals. Allows two sources to create a multitone signal.
Signal Coupler 	RF signal coupler for incident waves. Includes incident signal coupler and reflected signal coupler. Required for s-parameter measurements.
Step Attenuator	Attenuator in both the source and receive signal path for moderate (between coarse and fine) level adjustments.





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UltraWave 12G Theory of Operation : UltraWave 12G Logical Instruments and Hardware Functions

UltraWave 12G Logical Instruments and Hardware Functions

Any UltraWave 12G logical instrument or any valid combination of instruments operating simultaneously can be represented as a group of functions and switch settings on the UltraWave 12G Subsystem Function Block Diagram.

The following topics describe the configuration of the UltraWave 12G subsystem for the operation of each logical instrument:

- [UltraWave 12G MWModulatedSource Theory of Operation](#)
- [About the UltraWave 12G MWMultiToneSource](#)
- [About the UltraWave 12G MWNoiseSource](#)
- [About the UltraWave 12G MWReceiver](#)
- [UltraWave 12G MWSource Theory of Operation](#)
- [UltraWave 12G VNA Theory Of Operation](#)

The topics also provide information on hardware functions during operation of the specific logical instrument.

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UltraWave 12G Theory of Operation : User View of UltraWave 12G SMPM Connector Blocks

User View of UltraWave 12G SMPM Connector Blocks

This section shows the UltraWave 12G subsystem SMPM connector locations as seen from the user perspective (looking at the top, or device side, of the DIB/HIB/PIB and toward the test system). Two boards in the UltraWave 12G subsystem have SMPM connectors on sidewalk 1:

- | | |
|---|---|
| • | UltraWave 12G Measure Board |
| • | UltraWave 12G Synthesizer Reference Board |

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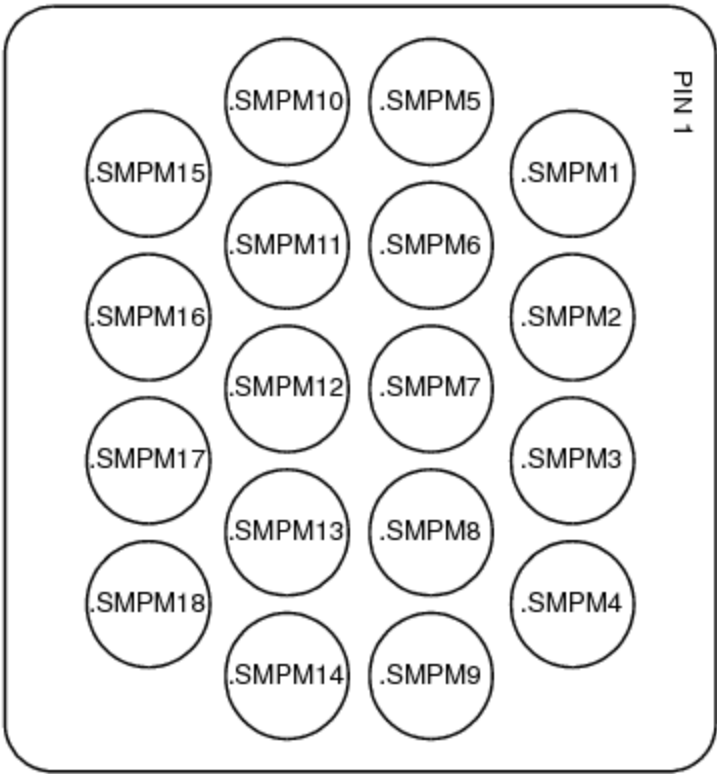
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UltraWave 12G Theory of Operation : User View of UltraWave 12G SMPM Connector Blocks :
UltraWave 12G Measure Board

UltraWave 12G Measure Board

User View



User View

Center of DIB

User View



UltraWave 12G Measure Board SMPM Connectors on Sidewalk 1 (User View)

		
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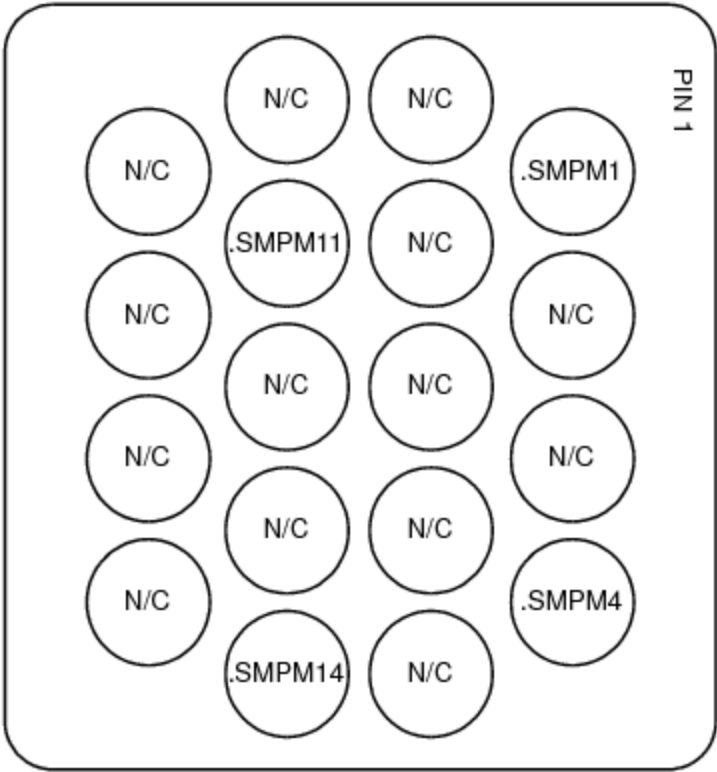
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UltraWave 12G Theory of Operation : User View of UltraWave 12G SMPM Connector Blocks :
UltraWave 12G Synthesizer Reference Board

UltraWave 12G Synthesizer Reference Board
User View



User View **Center of DIB** **User View**



UltraWave 12G Synthesizer Reference Board SMPM Connectors on Sidewalk 1 (User View)

These connectors are for DUT LO signals. For Reference (DUT) Clock connectors, see [User View of UltraWave 12G DIB Pad Patterns](#).

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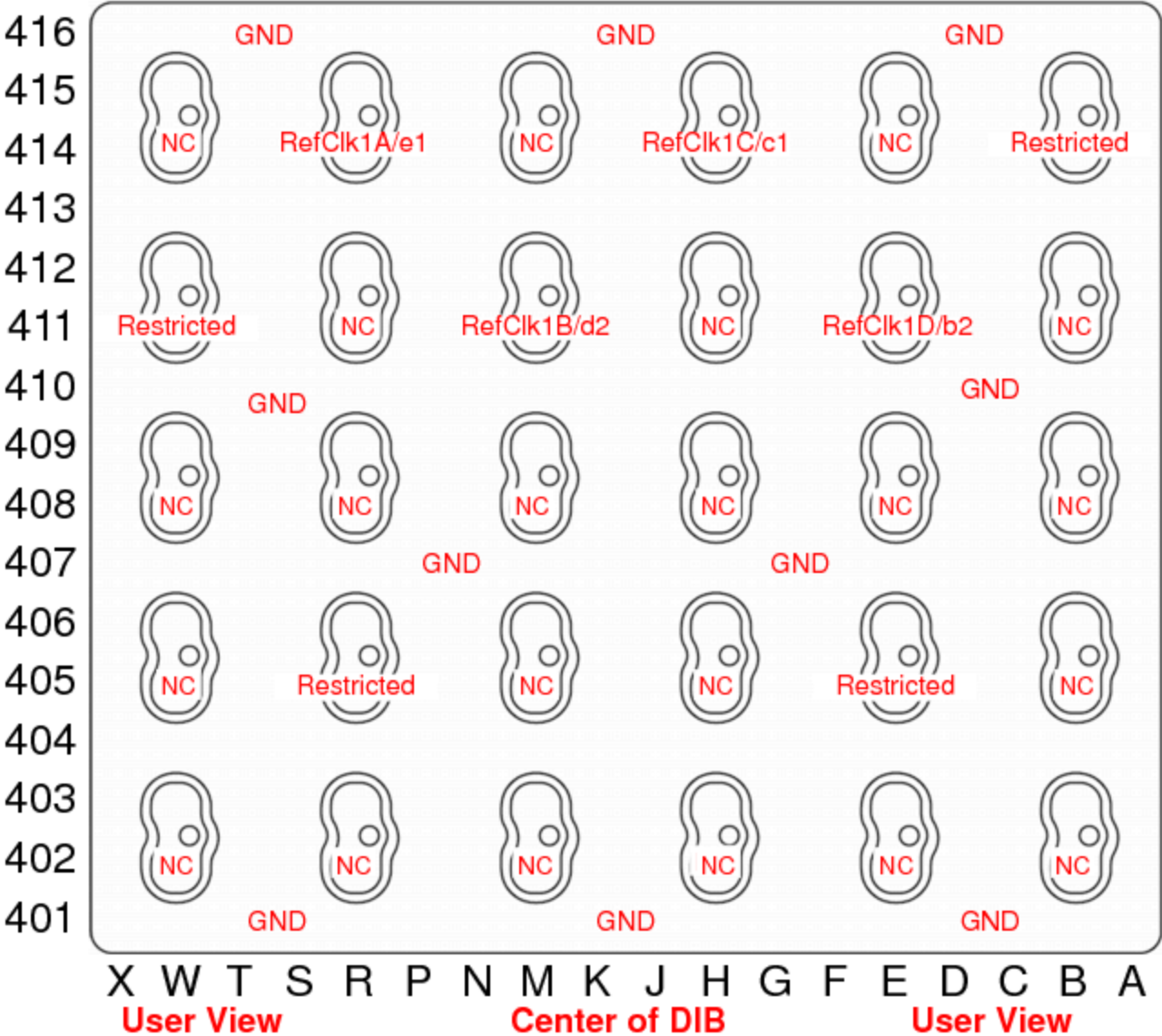
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UltraWave 12G Theory of Operation : User View of UltraWave 12G DIB Pad Patterns

User View of UltraWave 12G DIB Pad Patterns

This section shows the UltraWave 12G subsystem DIB pad connector locations as seen from the user perspective (looking at the top, or device side, of the DIB/HIB/PIB and toward the test system). The MW Synthesizer Reference board is the only board in the UltraWave 12G subsystem that has DIB pad connectors on sidewalk 4.

UltraWave 12G Synthesizer Reference Board - Ref Clock



UltraWave Synthesizer Reference Clock Pad Pattern for Sidewalk 4 (User View)

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UltraWave 12G Theory of Operation : UltraWave 12G Signal Descriptions

UltraWave 12G Signal Descriptions

The tables describe the SMPM connector names (or DIB pad names), the MW signal names, and the UltraWave 12G subsystem functions, where x is the corresponding test head slot. The MW Measure board and the Synthesizer Reference board have separate tables. The SMPM names are the same as the MW channel names in debug displays and the Pogo names in the channel map.

For the connector layout on the DIB interface, see:

- User View of UltraWave 12G SMPM Connector Blocks
- User View of UltraWave 12G DIB Pad Patterns

The tables for the two boards are:

- MW Measure Board Signal Connections
- MW Synthesizer Reference Board DUT LO Signal Connections
- MW Synthesizer Reference Board Ref Clock Signal Connections

MW Measure Board Signal Connections

MW Measure Board SMPM Connector Description

Channel Map Signal Name	Channel Map Pogo Name ¹	Function
x.MW1A	x.SMPM1	Source/Receive port A on port group 1.
x.MW1B	x.SMPM2	Source/Receive port B on port group 1.
x.MW1C	x.SMPM3	Source/Receive port C on port group 1.
x.MW1D	x.SMPM4	Source/Receive port D on port group 1.
x.MW2A	x.SMPM5	Source/Receive port A on port group 2.
x.MW2B	x.SMPM6	Source/Receive port B on port group 2.
x.MW2C	x.SMPM7	Source/Receive port C on port group 2.
x.MW2D	x.SMPM8	Source/Receive port D on port group 2.
N/A	x.SMPM9	Auxiliary I/F output of receiver for port groups 1 and 3.
N/A	x.SMPM10	Auxiliary I/F output of receiver for port groups 2 and 4.
x.MW3A	x.SMPM11	Source/Receive port A on port group 3.
x.MW3B	x.SMPM12	Source/Receive port B on port group 3.
x.MW3C	x.SMPM13	Source/Receive port C on port group 3.
x.MW3D	x.SMPM14	Source/Receive port D on port group 3.
x.MW4A	x.SMPM15	Source/Receive port A on port group 4.
x.MW4B	x.SMPM16	Source/Receive port B on port group 4.
x.MW4C	x.SMPM17	Source/Receive port group C on Mux 4.
x.MW4D	x.SMPM18	Source/Receive port group D on Mux 4.

¹ The sidewalk 1 connector block uses the same SMPM Pogo name to identify the signal location at the DIB.

MW Synthesizer Reference Board DUT LO Signal Connections

MW Synthesizer Reference Board SMPM Connector Description

Channel Map Signal Name	Channel Map Pogo Name ¹	Function
x.LO1A	x.SMPM1	DUT LO output A.

N/A	N/A	None.
N/A	N/A	None.
x.LO1B	x.SMPM4	DUT LO output B.
N/A	N/A	None.
N/A	N/A	None.
N/A	N/A	None.
N/A	N/A	None.
N/A	N/A	None.
N/A	N/A	None.
x.LO1C	x.SMPM11	DUT LO output C.
N/A	N/A	None.
N/A	N/A	None.
x.LO1D	x.SMPM14	DUT LO output D.
N/A	N/A	None.
N/A	N/A	None.
N/A	N/A	None.
N/A	N/A	None.

1 The sidewalk 1 connector block uses the same SMPM Pogo name to identify the signal location at the DIB.

MW Synthesizer Reference Board Ref Clock Signal Connections

MW Synthesizer Reference Board DIB Pad Connector Description

Channel Map Signal Name	Channel Map Pogo Name	DIB Sidewalk 4 Location	Function
x.RefClk1A	x.e1	R414-R415	Reference (DUT) clock output A.
x.RefClk1B	x.d2	M411-M412	Reference (DUT) clock output B.
x.RefClk1C	x.c1	H414-H415	Reference (DUT) clock output C.
x.RefClk1D	x.b2	E411-E412	Reference (DUT) clock output D.

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About the UltraWave 12G Subsystem

About the UltraWave 12G Subsystem

The UltraWave 12G subsystem provides a test instrument for sourcing, measuring, and analyzing RF signals in the frequency range from 50MHz to 12GHz. The UltraWave 12G subsystem is made up of instrument function modules on boards and modules in the test head, and instrument software. IG-XL software provides MW logical instruments that use specific combinations of function modules to source and capture signals for RF measurements. The logical instrument interfaces simplify programming of hardware resources.

Testing applications include mobile phone transceivers, wireless LAN chipsets, RF amplifiers, TV tuners, and IF devices. Typical tests are RF power, gain, IP3, noise figure, and phase noise. This section includes the following topics:

- | | |
|---|---|
| • | Features |
| • | Safety Information |
| • | Key Terms |
| • | UltraWave 12G Software Licensing Requirements |

UltraWave12G subsystem information includes the following sections:

- | | |
|---|--|
| • | UltraWave 12G Theory of Operation |
| • | UltraWave 12G Instrument Calibration |

- [UltraWave 12G Instrument Programming](#)
- [UltraWave 12G Displays](#)
- [UltraWave 12G Examples](#)

Related Topics

- [UltraWave 12G Specifications](#)
- [UltraWave 12G DIB Design Guidelines](#)
- The UltraWave 12G subsystem supports all UltraWave 12G MW logical instruments. For more information, see:
 - [About the UltraWave 12G MWModulatedSource](#)
 - [About the UltraWave 12G MWMultiToneSource](#)
 - [About the UltraWave 12G MWNoiseSource](#)
 - [About the UltraWave 12G MWReceiver](#)
 - [About the UltraWave 12G MWSource](#)
 - [About the UltraWave 12G VNA](#)

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